<<<<<<<<<<<# exif-turbo User Manual

exif-turbo lets you scan image folders, build a searchable index of all EXIF metadata, and instantly find any photo by camera model, lens, date, location, or any other tag — across thousands of images. Video files (MP4, MOV, AVI, MKV, WMV, M4V, MTS, M2TS, 3GP, WebM, FLV) are indexed alongside still images, with thumbnails and previews extracted from the embedded thumbnail or a frame at 1/3 of the duration.

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1. Requirements

ExifTool

exif-turbo requires **ExifTool** on your `PATH` to extract metadata from images.

Platform	Install

Windows (MSI)	ExifTool is **bundled with the MSI installer** — no separate download needed. A system-wide `exiftool.exe` on your `PATH` takes priority if present.
Windows (source)	Download the standalone `.exe` from exiftool.org, rename to `exiftool.exe`, place in a folder on your `PATH`
macOS	`brew install exiftool` (if Homebrew is not installed yet: `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install`
Linux	`sudo apt install libimage-exiftool-perl`

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2. Installation

Windows installer (recommended)

Download `exif-turbo-<version>-windows.msi` from the Releases page. The installer adds an entry to **Start Menu** and puts `exif-turbo` on your `PATH`. ExifTool is **included in the MSI** — you do not need to download it separately. If you already have ExifTool installed system-wide, that version takes priority.

The feature-selection screen lists these components individually. FFmpeg and ImageMagick are separate entries, not grouped under derivative tooling:

Component	Purpose	Default
exif-turbo	The application	Selected
ExifTool (bundled, recommended)	Reading and writing image metadata	Selected
FFmpeg	Extracting video frames for derivative export; includes FFprobe and shared libraries	Selected
ImageMagick	Converting and encoding derivative images	Selected

You can deselect FFmpeg and ImageMagick independently. Converted image exports need ImageMagick; video-frame exports also need FFmpeg. Original, byte-for-byte copies do not use either converter. Installed system tools on `PATH` take priority over bundled copies. The bundled tools are not added to the system `PATH`.

Each selected tool includes its license and third-party notices. ImageMagick also includes its upstream software bill of materials; FFmpeg includes matching source, build scripts, configuration details, and checksums. These materials are installed in `media-tools\licenses\` under the application directory, with additional upstream notices alongside each tool. The application's `licenses` folder includes the complete third-party package and license inventory.

macOS installer

Download `exif-turbo-<version>-macos-arm64.dmg` for Apple silicon from the Releases page, open it, and drag **exif-turbo.app** into your **Applications** folder.

Linux package

Download `exif-turbo-<version>-linux-amd64.deb` or `exif-turbo-<version>-linux-arm64.deb` (Debian/Ubuntu), or `exif-turbo-<version>-linux-x86_64.rpm` (Fedora/openSUSE), from the Releases page and install it with your package manager:

```
# Debian / Ubuntu
sudo apt install ./exif-turbo-<version>-linux-amd64.deb

# Fedora / openSUSE
sudo dnf install ./exif-turbo-<version>-linux-x86_64.rpm
```

The package installs the application to `/opt/exif-turbo/`, registers a desktop launcher, and creates a `/usr/bin/exif-turbo` symlink.

From source

```
pip install -e .
```

Command-line options

```
exif-turbo [--db NAME] [--version]
```

Option	Description

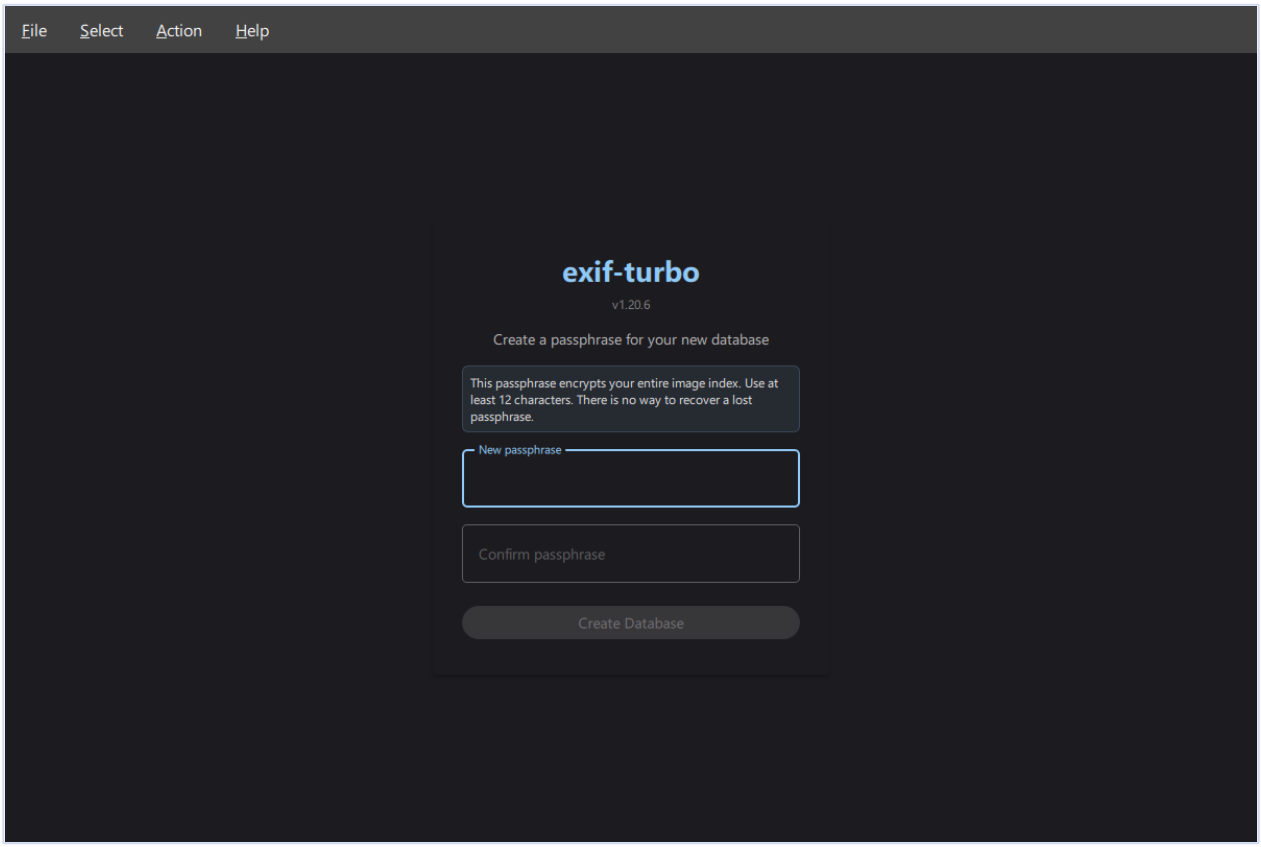
`--db NAME`	Open (or create) a named database instead of the default one. The database is always stored in `~/.exif-turbo/data/<NAME>/<NAME>.db`. Useful for keeping separate libraries — e.g. `exif-turbo --db work` and `exif-turbo --db holidays`.
`--version`	Print the installed exif-turbo version and exit.

If `--db` is omitted, the default database `~/.exif-turbo/data/index/index.db` is used.

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3. First Launch — Unlocking the Database

When you start exif-turbo you are greeted by the **lock screen**:



First-time setup — creating a new database

If this is the first time you have launched exif-turbo (or you are using a new named database), the lock screen shows a **New passphrase** field, a **Confirm passphrase** field, and a **Create Database** button. Choose a strong passphrase of at least 12 characters that mixes letters, numbers, and

symbols — it encrypts your entire image index and **cannot be recovered if lost**.

What the password protects

For password-protected databases, encryption at rest covers the SQLCipher database (indexed metadata, image paths, and cached tags), thumbnail and preview caches, and the AI index (vectors, image-path mappings, and model metadata). This does not encrypt the original images or videos, adjacent tagging sidecars, exported metadata, saved previews, copied originals, or tagged derivatives. Settings files and diagnostic output are not encrypted and may reveal paths or other activity details. Clipboard contents are also outside this protection.

The AI index is stored as a single authenticated AES-256-GCM snapshot, `ai_index.faiss.enc`, with a separate password-wrapped key in `ai_index.faiss.keys/.thumb_key`. Keep the key directory with the snapshot when backing up or moving a database. Clearing thumbnails does not delete this key. Snapshot encryption and verification process bounded chunks in memory; the FAISS index and serialized vector data still require memory. AI saves do not stage plaintext vectors or mappings on disk. File names, sizes, and timestamps remain visible to someone who can inspect the storage directory.

Existing plaintext AI files (`ai_index.faiss`, `ai_id_map.json`, and `ai_index_meta.json`) are migrated on the next successful password unlock, even when AI search is disabled. The encrypted copy is verified before the plaintext files and known legacy temporary files are removed. This can take time and needs additional free disk space and memory for a large index. Failed migration leaves the originals available for retry. Removing files is not secure erasure: older backups, filesystem snapshots, and recoverable disk blocks may still contain plaintext data. Earlier app versions cannot read the new encrypted snapshot; do not run an older version against a migrated database.

Missing keys, wrong AI passwords, and failed authentication are reported instead of silently falling back to plaintext or overwriting the encrypted snapshot during **AI Full Rescan**. Restore the snapshot and its matching key directory from a trusted backup if necessary. Incomplete or incompatible legacy data is preserved inside the encrypted snapshot and may still require **AI Full Rescan**.

While the app is unlocked, decrypted data exists in process memory. Closing the app stops workers and releases its password/key references, but Python and native libraries cannot guarantee secure memory erasure. Encryption does not protect against a compromised operating system, malware, screen capture, or swap/crash dumps. Explicitly enabled passwordless development databases and their caches are not encrypted.

Opening an existing database

Enter your password in the **Password** field and click **Unlock** (or press **Enter**). The same password is required every time you open the app.

Once unlocked, the **Search** tab opens and any previously indexed images are immediately available.

> **ExifTool not found:** If ExifTool is not installed or not on your `PATH` at the time of unlock, a **"ExifTool not found"** dialog appears automatically. It explains that indexing is disabled and provides a link to exiftool.org. Install ExifTool and restart the app to enable indexing. You can still search and browse existing index data while ExifTool is missing.

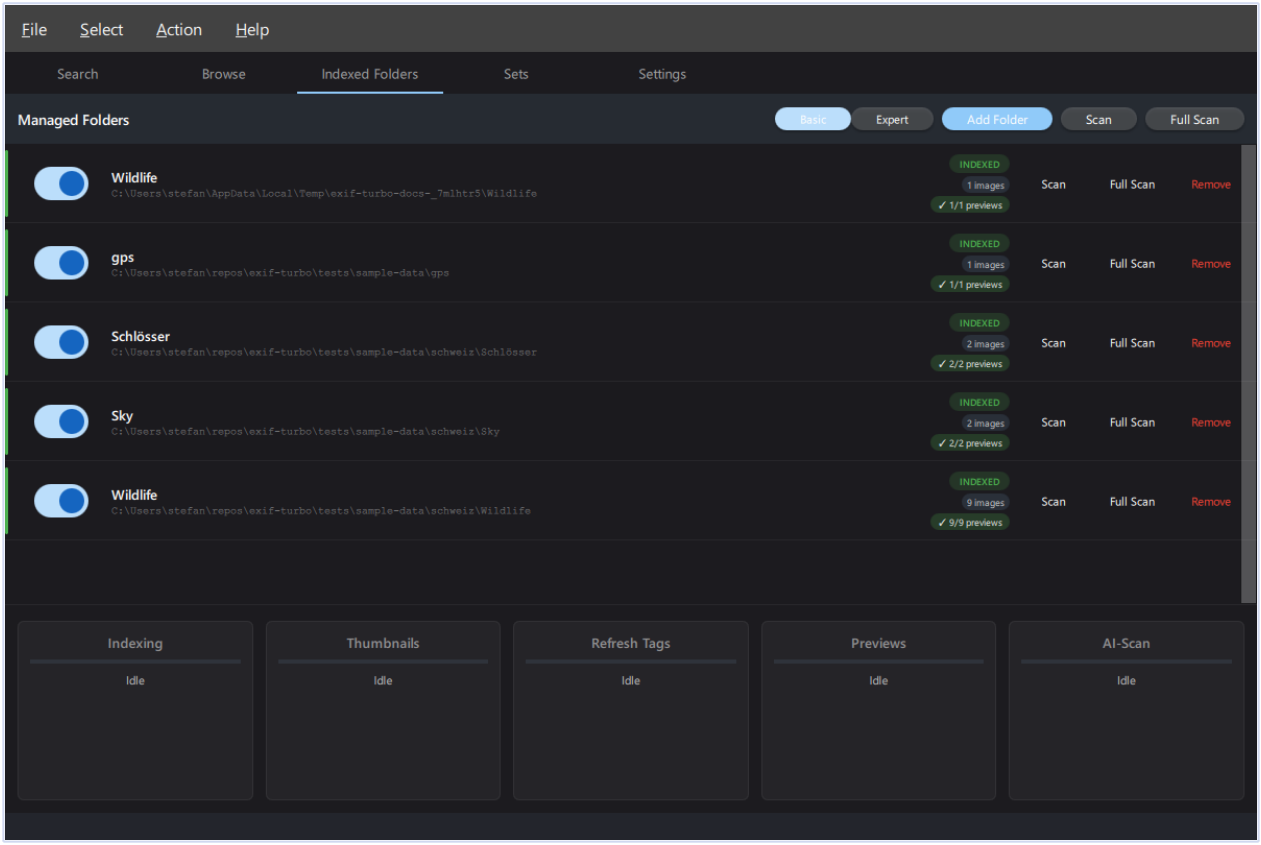
Help menu

The **Help** menu in the menu bar provides access to this user manual, the third-party open-source licence list, and the **About** dialog, which shows the application version, a brief description of the application, and the licence (MIT).

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4. Indexed Folders — Managing Your Library

Click the **Indexed Folders** tab to manage which directories are scanned.



Screenshot contains no third-party photographs.

Adding a folder

1. Click **Add Folder** in the **Managed Folders** header bar at the top of the tab.
2. Pick the directory in the file browser dialog.
3. The folder is immediately queued for scanning — its status changes to **QUEUED** then **SCANNING** once the worker starts.

The header bar provides a **Basic / Expert** mode selector. **Basic** is the default and keeps each folder row focused on the two complete workflows most people need. **Expert** exposes each operation separately. The selected mode applies to the Indexed Folders view only.

The header **Scan** and **Full Scan** buttons run the corresponding complete workflow for every enabled folder, one folder at a time. The row buttons with the same labels apply only to that folder.

Starting an index scan

In Basic mode, click **Scan** next to a folder to incrementally re-index it, refresh its sidecar tags, build missing previews, and run an incremental AI scan when AI features are enabled. The folder status changes to **SCANNING** and the progress panel appears in the bottom-right corner of the tab.

Use **Full Scan** to force every file to be re-processed, refresh sidecar tags, clear and rebuild previews, and rebuild AI data when enabled. This is useful after updating ExifTool or if you suspect any derived data is out of date.

When rescanning a single folder, only the records belonging to that folder are updated or removed. Images indexed from other folders are not affected.

Folder statuses

Each folder row shows a coloured status badge, an image-count badge (e.g. "42 images") once it has been indexed, and — if previews have been rendered — a preview-cache badge ("✓ 42/42 previews"). The preview badge is **green** when every image in the folder has a cached preview and **amber** when only some do; hover it to see the exact ratio.

Status	Meaning
NEW	Added but never scanned
QUEUED	Waiting in the scan queue
SCANNING	Currently being indexed
INDEXED	Last scan completed successfully
MISSING	Folder path no longer exists on disk

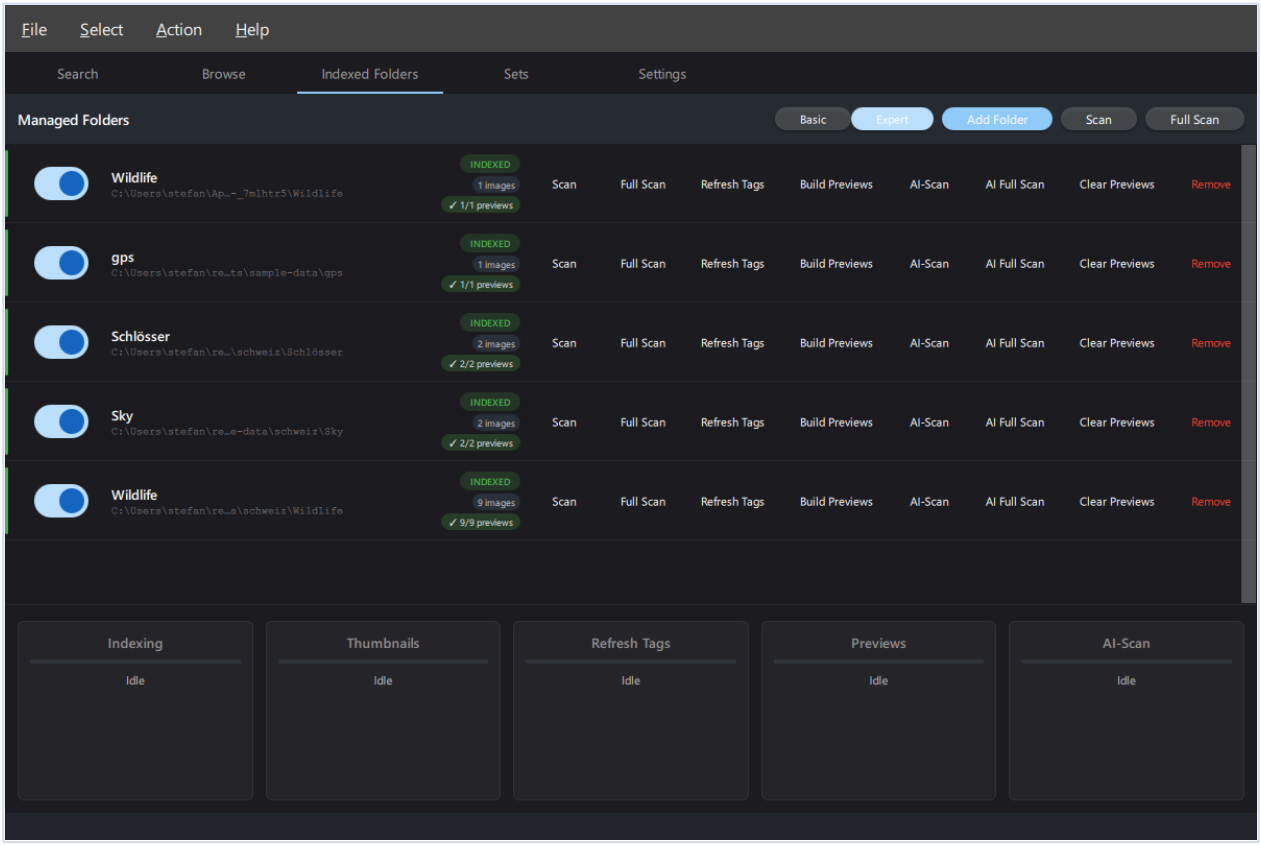
ERROR	Last scan ended with an error
DISABLED	Excluded from search results

Disabling / enabling a folder

Toggle the **Enabled** switch to exclude a folder from search results without deleting it or its index data. Hovering the switch shows the tooltip *"Folder is included in search results"* (when on) or *"Folder is excluded from search results"* (when off).

Expert actions

Switch to **Expert** to expose the individual operations on each row:



Screenshot contains no third-party photographs.

Button	Action
Scan	Incrementally re-index this folder (only files whose modification time changed).
Full Scan	Force re-extract EXIF for every file in this folder.

Refresh Tags	Force re-read sidecar files for this folder's indexed images without re-extracting EXIF or rebuilding previews. Newly added and changed sidecars update cached custom tags and descriptions, including search text; deleted sidecars remove both from the cache. Malformed sidecars are left untouched and reported as errors. The operation appears in the persistent activity panel and can be canceled there.
Build Previews	Render preview-cache JPEGs for every image in this folder. While the build is running on this folder the same button reads **Cancel Previews**. Disabled while another folder's preview build is in progress.
AI-Scan	Build missing CLIP vector embeddings for this folder only (incremental semantic-index build). While running, the same button reads **Cancel AI-Scan**. Visible only when AI features are enabled in Settings.
AI Full Scan	Rebuild every CLIP vector embedding for this folder from scratch. While running, the same button reads **Cancel AI Full Scan**. Visible only when AI features are enabled.
Clear Previews	Delete all cached previews for this folder. Hidden (kept invisible for layout alignment) when nothing is cached. A confirmation dialog asks *"Delete N cached preview(s) for \"\"? Thumbnails are unaffected."*
Remove	Remove the folder and delete all its indexed images. A **Remove Folder** confirmation dialog asks before deletion. The original files on disk are not touched. The removal then runs on a background worker behind the modal **bulk-op progress overlay** (see section 10), which reports each sub-step — *"Clearing preview cache…"* (cancelable, with an `X / Y` count) followed by *"Deleting index entries…"*.

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5. Indexing Progress

The bottom of the **Indexed Folders** tab contains five persistent activity columns: **Indexing**, **Thumbnails**, **Refresh Tags**, **Previews**, and **AI-Scan**. Each column shows its own operation independently. When a column is idle, its bar is empty and its status reads **Idle**.

- **"Indexing folder N of M"** / **"Indexing"** — file-indexing phase. Shown while the indexer is processing folders from the queue.
- **"Thumbnails"** — thumbnail-cache phase, started automatically after indexing finishes.
- **"Refresh Tags"** — sidecar-tag refresh phase, started directly in Expert mode or after indexing in a Basic workflow. It runs in this activity column without opening the modal bulk-operation overlay.
- **"Previews"** — preview-cache phase, started by the **Build Previews** button on a folder row.
- **"AI-Scan"** — CLIP image-vector phase, started directly in Expert mode or as the final phase of a Basic workflow.

Each active column contains:

- **Progress bar** — indeterminate while the total is still being computed, then a percentage once it is known.
- **Count label** — **"Scanning for images…"** at the start of indexing discovery; **"N indexed, scanning…"** while indexing has started but the total file count is not yet known; **"Preparing…"** for the thumbnail and preview phases before the total is known; then `n / total files` (indexing) or `n / total images` (thumbnails and previews).
- **Current file** — name of the file being processed.
- **Cancel button** — stops that operation. While an indexing or thumbnail cancel is in flight the label changes to **"Canceling…"** and the button is disabled until the worker has stopped.

While AI scanning is active, the affected folder row also shows a spinner and an **AI-Scan** or **AI Full Scan** badge. Its normal folder status returns when the operation stops.

Across **all tabs**, folder-operation messages in the **status bar** begin with the indexed folder name. While indexing, refreshing tags, building thumbnails or previews, or running AI-Scan, that folder name pulses in the accent colour so the active folder remains visible even when you are working in Search or Browse. There is no separate activity dot or redundant **Indexing…** label. The status bar also shows brief event messages to its right (such as "Indexed 42 images" after a scan completes).

Self-healing cache cleanup

At the end of every successful folder index run, exif-turbo runs a quick garbage-collection pass over the thumbnail and preview cache directories. Any cached file whose source image is no longer present in the index (because the file was deleted, moved, or renamed since the last scan) is removed. The status bar shows **“Cleaning up cache…”** while the sweep runs. This keeps the cache compact and self-heals across crashes or external file deletions, so you never need to

manually clear the cache directory.

Pause and resume

If you close the application while indexing is in progress, the current folder is saved as **QUEUED**. The next time you open exif-turbo and unlock the database the scan queue is automatically restored and indexing resumes where it left off.

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6. Searching

The **Search** tab is the main way to find photos:

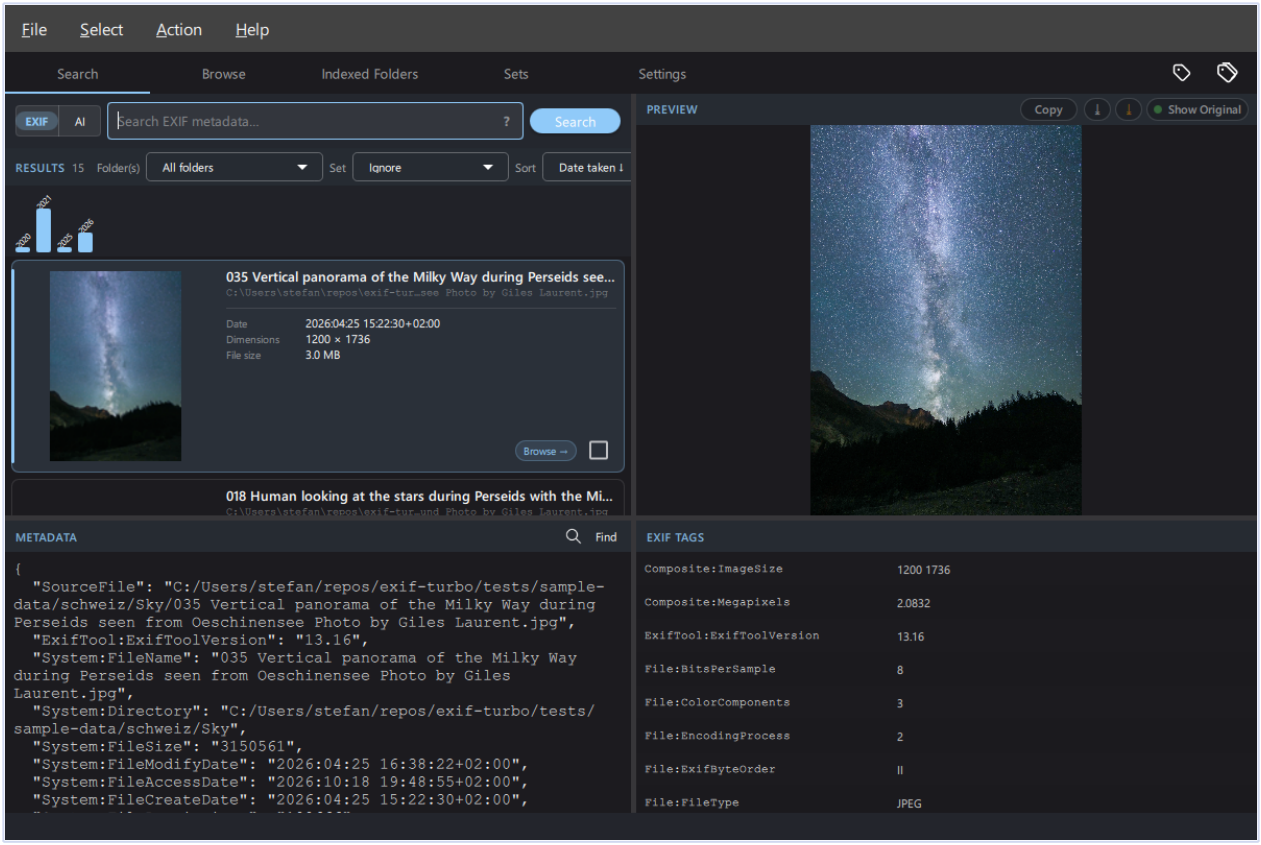


Photo: Wild Alpine Ibex, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

Running a search

Type any word or phrase into the **Search EXIF metadata…** bar and press **Enter** (or click **Search**). exif-turbo performs a full-text search across every metadata field — camera make and model, lens name, date, GPS coordinates, keywords, copyright, and more.

Press **Enter** with an empty search bar to show all indexed images.

A **?** button at the right edge of the search field shows a syntax cheat-sheet when hovered. It includes full-text and structured search examples. You do not need to click it — the tooltip appears automatically on hover.

When the search field contains text a **×** button appears to its left. Clicking it clears the field and immediately shows all images (equivalent to pressing **Enter** with an empty bar).

While a search is running, a semi-transparent grey overlay dims the entire UI and the cursor changes to a busy indicator. The overlay clears automatically when results are ready.

AI semantic search (EXIF/AI toggle)

When **AI Features** are enabled in **Settings**, the search bar shows an **EXIF / AI** toggle. Switch to **AI** mode to search by meaning instead of exact metadata terms.

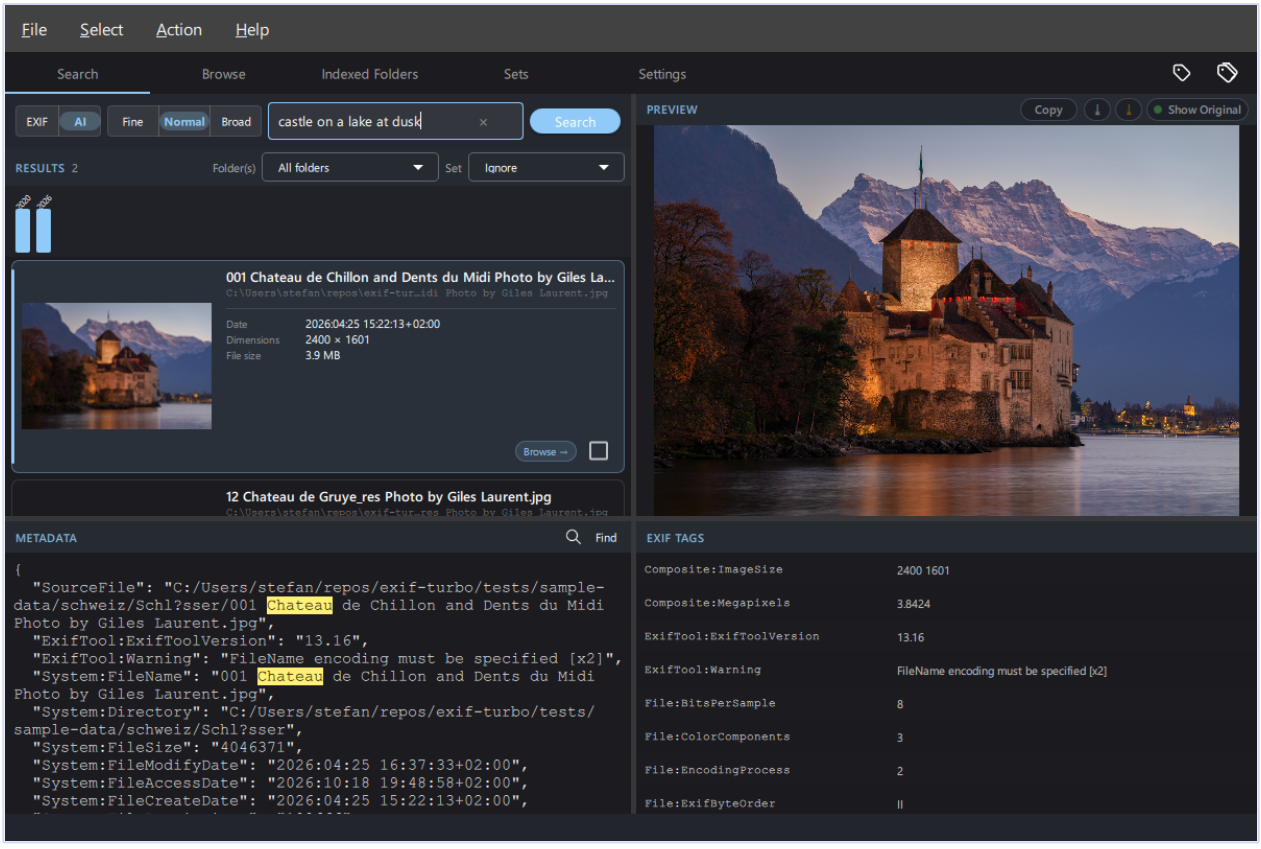


Photo: Chateau de Chillon and Dents du Midi, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

In AI mode:

1. Enter a natural-language query such as *"golden eagle over mountain lake"*.
2. Press **Enter** (or click **Search**) to run CLIP semantic retrieval.
3. Choose a precision level:
4. **Fine**: strictest matches (score >= 0.22)

5. **Normal**: balanced default (score >= 0.20)
6. **Broad**: most permissive (score >= 0.18)

AI search requires CLIP vectors to exist for the target images. Build them with **AI-Scan** (or **AI Full Scan**) in the **Indexed Folders** tab.

AI vector indexes are tied to the model that created them. If the application reports missing or incompatible AI index metadata after an upgrade, run **AI Full Scan** before using AI search.

Note for macOS Intel users: AI features are unavailable on macOS Intel (x86_64) targets and are shown disabled in **Settings**. This is due to PyTorch support for Python 3.13+ on that platform.

Filtering by format

When results contain more than one file format, a row of format chips appears below the search bar. Click a chip to show only that format:

```
All   CR2 · 1459   JPG · 563   TIF · 113   PNG · 2
```

Click **All** to return to unfiltered results.

Filtering by folder

When you have two or more indexed folders enabled, a **Folder(s)** dropdown appears in the RESULTS header bar, to the left of the **Sort** control. It is hidden when only one folder is indexed.

The button label reflects the current filter state:

Label	Meaning
All folders	No folder filter — results span all enabled folders
Folder name	Exactly one folder is selected
N folders	N individual folders are selected

Click **Folder(s)** to open the selection popup. Tick **All folders** to clear any active filter; tick one or more folder names to restrict results to those folders only. Multiple folders can be selected simultaneously. Hovering over a folder name shows its full path as a tooltip.

> **Drive roots on Windows** — when an entire drive (e.g. `C:\`) is added as an indexed folder it appears in the dropdown with a friendly label such as **"OS (C:)"** (volume name + drive letter), rather than a blank entry.

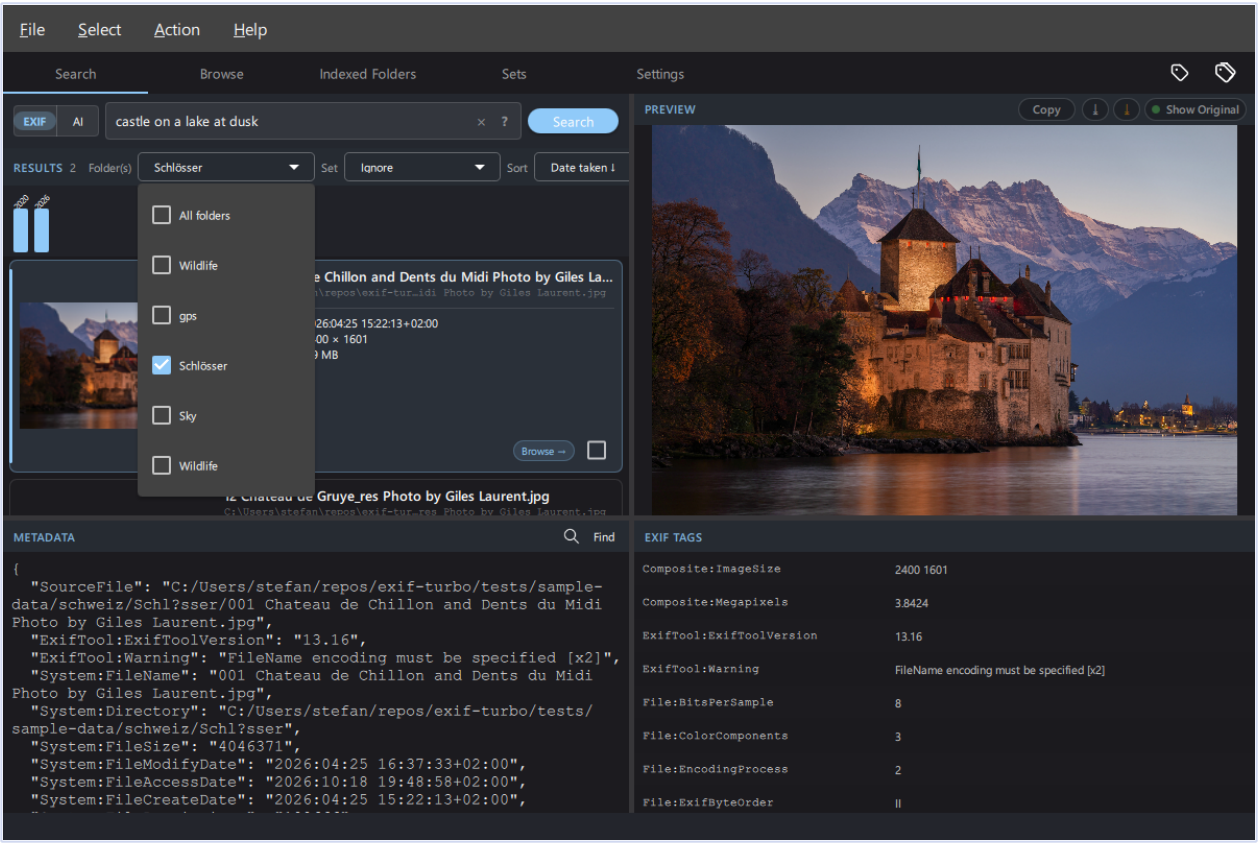


Photo: Chateau de Chillon and Dents du Midi, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

The selection chip

Whenever at least one image is marked (or the marked-only filter is active), a **selection chip** appears in the RESULTS header bar next to the **Folder(s)** dropdown. It shows the current marked count:

Label	Meaning
■ `N`	Every marked image is in the current results
■ `here/total`	Only some marked images match the current filters (e.g. ■ `12/87`)

Click the chip to toggle the **marked-only filter** — the tooltip switches between *"Show only selected images"* and *"Show all results"*. While the filter is active the chip stays highlighted so you always know which view you are looking at.

Sorting results

Use the **Sort** dropdown at the top-right of the results panel:

Option	Description
Date taken ↓	EXIF capture date, newest first (default)
Date taken ↑	EXIF capture date, oldest first
Name A→Z	Filename ascending
Name Z→A	Filename descending
Path A→Z	Full path ascending
Path Z→A	Full path descending
Largest	File size, largest first
Smallest	File size, smallest first

Date taken ↓/↑ sorts by the capture timestamp stored during indexing. exif-turbo resolves this timestamp from a prioritised chain: primary EXIF fields (`DateTimeOriginal`, `CreateDate`) are tried first; if none are present, XMP, IPTC, and QuickTime creation fields are consulted; as a last resort the file-system creation time (macOS/Windows) or modification time (Linux) is used. Infrastructure metadata such as ICC colour-profile dates is never used. Images with no resolvable date appear at the end of the list in both directions.

The chosen sort order is remembered per database and restored automatically the next time you open the application.

Filtering by capture year

Whenever at least one indexed image has a known capture date a **year histogram** appears below the format chips in the Search tab. Each bar represents one calendar year; its height is proportional to the number of matching images taken in that year relative to the busiest year. A small spinner appears in this area while the timeline counts are loading or refreshing.

For non-empty AI queries, the active folder, format, and capture-date filters determine eligibility before the 2,000-image display limit is applied. Images from other years cannot crowd out eligible matches. Results remain ordered by semantic similarity, with multiple views pooled into one result per image. The AI timeline counts all threshold-qualified images in the active folder and format scope, independently of the selected dates and display limit. An empty AI query continues to show all images in scope, subject to the active filters.

Action	Effect

Click a bar	Filter results to images captured in that year
Shift-click a different bar	Extend the selection to cover the range between the two bars
Click the active bar again	Clear the year filter
× chip (right side of histogram)	Clear the year filter

A tooltip on each bar shows the year and image count. When a filter is already active, hovering a bar that is not the sole selected year shows "Shift-click to extend range".

Images that have no resolvable capture date are excluded from results while a year filter is active.

Loading more results

Results are loaded in batches as you scroll. When you reach the bottom of the result list the next batch is fetched automatically. The total match count is shown next to the **RESULTS** badge in the panel header.

Opening images from results

Single-click a result card to select it and load the preview and metadata panels.

Double-click a result card to open the file or folder in your system's default application: - Double-clicking the **thumbnail** (left side of the card) opens the image in your default image viewer. - Double-clicking the **info area** (right side of the card) opens the parent folder in the system file manager. On Windows, Explorer opens with the file highlighted; on macOS and Linux the parent folder is opened.

If the original file cannot be reached — for example when the folder that contains it is not currently mounted or attached — nothing is opened and a **red warning** appears in the status bar at the bottom of the window naming the indexed folder you need to (re)attach or mount, e.g. *"Original data source not attached: D:\Photos — attach or mount this folder to open files."*

Search examples

Query	What it finds
`Canon EOS R5`	All images shot with a Canon EOS R5
`"golden eagle"`	Images with that exact phrase in indexed text

`Switzerland 2024`	Images containing both terms in indexed text
`Canon OR Nikon`	Images containing either camera make
`Fuji*`	Images with a token beginning with `Fuji`

Advanced query syntax

The search box accepts the full **SQLite FTS5** query language:

Syntax	Example	What it does
`term`	`Canon`	Keyword anywhere in metadata
`"exact phrase"`	`"red deer"`	Terms must appear adjacent and in order
`term1 AND term2`	`Canon AND 50mm`	Both terms must be present
`term1 OR term2`	`Canon OR Nikon`	Either term must be present
`term1 NOT term2`	`"red deer" NOT captive`	Phrase present, excluded term absent
`prefix*`	`Fuji*`	Matches any token starting with the prefix

Multiple terms without an operator (`Canon 50mm`) are treated as an implicit AND.

ExifTool stores metadata with group-prefixed keys such as `GPS:GPSLatitude` or `ExifIFD:FocalLength`. You can type these directly in the search box — the colon is treated as a word separator, so `GPS:GPSLatitude` becomes an implicit AND of `GPS` and `GPSLatitude`.

Exact tag, description, and text filters

Use structured filters when you need exact tag membership or want to search non-tag text separately from tags:

Syntax	Example	What it finds
`tags:=`	`tags:="red deer, landscape"`	Images with both exact effective tags

`embedded-tags:=`	`embedded-tags:="landscape"`	Images with that non-excluded embedded keyword
`custom-tags:=`	`custom-tags:="summer holiday"`	Images with that custom sidecar tag
`description:=`	`description:="sunset"`	Descriptions containing `sunset` anywhere
`text:=`	`text:="golden eagle"`	Images with that phrase in non-tag text
`text-not:=`	`text-not:="captive"`	Excludes the phrase from non-tag text

Tag values are matched case-insensitively as complete values, not as words or substrings. Multiple values are comma-separated and all must be present. The default `tags:=` filter searches the effective collection: embedded keywords not excluded for that image, plus custom sidecar tags. Excluding all embedded keywords does not exclude custom tags.

Tag values are matched case-insensitively as complete values, not as words or substrings. Commas separate multiple tags, and every listed value must be present. The default `tags:=` filter searches the effective collection: embedded keywords not excluded for that image, plus custom sidecar tags. Excluding all embedded keywords does not exclude custom tags.

Description filters perform case-insensitive substring matching across both embedded and custom descriptions. Unlike exact tags, they can match inside a word; for example, `description:="sun"` can match *sunset*.

`text:=` and `text-not:=` use FTS phrase matching on filenames, non-keyword embedded metadata, and custom descriptions. Embedded IPTC/XMP keyword values and custom sidecar tags are excluded from this text collection. These operators match phrases, not arbitrary substrings.

Combining expressions and filters

The ordinary full-text part can use the FTS operators above, and can be combined with structured filters. Each structured filter is ANDed with the ordinary full-text result and with every other structured filter. For example:

Query	Meaning
`(Canon OR Nikon) tags:="wildlife"`	Canon or Nikon images that have the exact effective tag `wildlife`

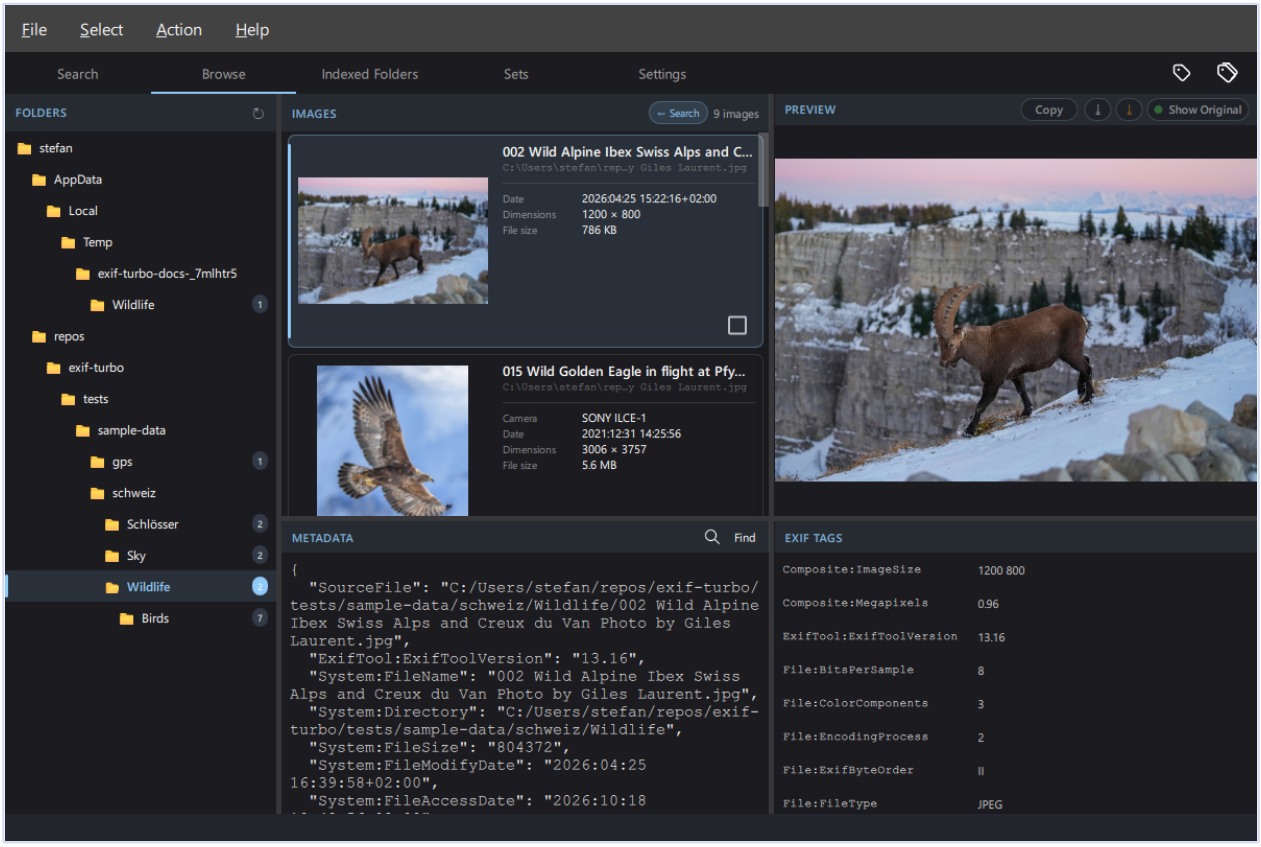
`"red deer" NOT captive tags:="wildlife, landscape"`	The phrase without `captive`, plus both exact effective tags
`tags:="red deer" AND description:="golden"`	The exact tag and a description containing `golden`
`tags:="red deer" text:="alpine lake" text-not:="zoo"`	The exact tag, the phrase in non-tag text, and no `zoo` phrase in non-tag text

When filters are separated by spaces, they already mean AND; writing `AND` between structured filters is optional. `OR` applies only inside the ordinary FTS expression. OR between structured filters (for example, either of two different `tags:=` filters) is not supported; a comma-separated tag list always requires every listed tag.

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7. Browsing by Folder

The **Browse** tab lets you navigate your library by folder hierarchy:



Photos: Wild Alpine Ibex and Wild Golden Eagle, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

The left panel shows all indexed folders as an indented list — sub-folders are indented under their parent. Each entry shows the folder name and a count of images inside it. Click any folder to show only its images in the centre panel. Click the highlighted folder again to deselect it and show all images.

The same thumbnail list and preview pane appear as in Search. **Single-click** an image to select it and load the preview. The preview supports the same zoom and pan gestures as in Search (scroll wheel, touchpad scroll, touchpad pinch-to-zoom, drag-to-pan, double-tap to reset — see Zooming and panning). **Double-click** an image to open it in your system's default viewer. (Unlike the Search tab, double-clicking in Browse always opens the image — there is no folder-open shortcut.)

The Browse tab shows the same **METADATA** and **EXIF TAGS** panels as the Search tab, including the GPS location bar. They appear in a split view below the image list and preview panel. The inline **Ctrl+F** find bar works on the Browse tab too — click **Find** in the METADATA panel header (or press **Ctrl+F** when the Browse tab is active) to search within the metadata text.

The Browse tab preview pane has the same toolbar pill buttons as the Search tab:

- **Copy** — copies the currently displayed preview image to the system clipboard. A brief toast notification confirms the action.
- **Save Preview As** (white ■) — opens a **Save File** dialog to save the preview as JPEG or PNG.
- **Save Original As** (orange ■) — copies the original source file byte-for-byte to a destination you choose.
- **Show Original / Show Preview** toggle — switches between the cached preview (green dot) and the full-resolution source file (orange dot). Only shown when the selected image has a cached preview.

While a full-resolution original is loading, a **"Loading original…"** spinner overlay appears over the preview area (only when **Show Original** is active).

Navigating the Browse image list

The image list takes keyboard focus as soon as a folder is selected. You can move through images without touching the mouse:

Key	Action
↓	Select the next image
↑	Select the previous image
`Page Down`	Jump one page forward
`Page Up`	Jump one page backward

A permanently visible vertical scrollbar (12 px) is always shown on the right edge of the list so you can drag it or click to jump to any position without first hovering the right edge.

Search-tab filter state is preserved

The Browse tab has its own independent navigation context. When you switch from Search to Browse and back, your active Search query, format chip, date range, sort order, extension filter, and folder filters are all restored exactly as you left them — Browse navigation never affects the Search state.

Switching back to the **Search** tab restores the search state exactly as you left it. Any image previously selected while browsing is not automatically highlighted in Search.

Jumping between Search and Browse

Each result card in the **Search** tab has a **Browse** → pill button in its bottom-right corner (to the left of the selection checkbox). Clicking it:

1. Switches to the **Browse** tab.
2. Navigates to the folder that contains that image.
3. Scrolls the Browse image list to that exact image and selects it.

The Browse tab image list header has a ← **Search** back button. Clicking it returns to the **Search** tab and restores the exact scroll position and previously-selected image — no need to re-enter the search query.

---

8. Image Sets

Image Sets are named collections of references to images in the current database. They do not copy, move, or modify image files. Membership stays fixed across searches and restarts until you change it or explicitly reapply the set's optional saved filter.

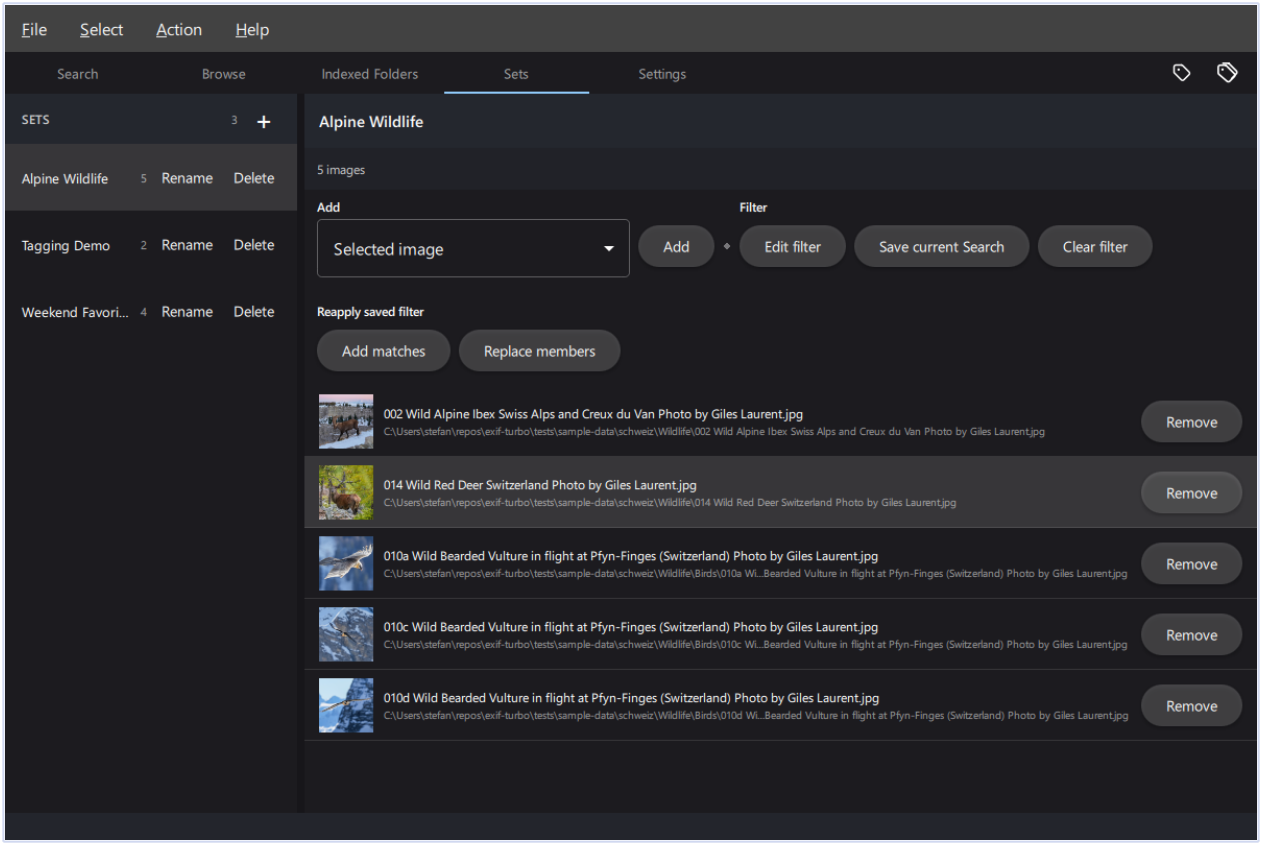


Photo: Wild Red Deer Switzerland, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

Create and manage sets

Open the **Sets** tab. Click **+** to create an empty set, then select it to view its members. Set names must be unique within the database, ignoring letter case. Use **Rename** or **Delete** on a set row to manage it. Deleting a set removes only its name, saved filter, and memberships; images, sidecars, and tags are not changed.

Use the **Add** scope to add the selected image, marked images, all current Search results, or every image in the current Browse folder. Search-result and folder operations include the complete scope, not only rows currently loaded or visible. After an AI search, **Current results** uses the exact AI result list, not a metadata search for the same words. Visiting Browse before opening Sets preserves that Search scope; use **Browse folder** to add the browsed folder instead. An empty AI result list adds no images. Click **Remove** on a member row to remove that image from the set; removing it does not delete the image from the database or disk. Repeating an add or remove is safe. Removing an image from the index also removes its set memberships.

Search by set

The **Set** filter in the Search toolbar controls which set members appear:

- **Ignore** applies no set filter, so Search can return any indexed image.

- **All sets** returns images belonging to at least one set.
- A named set returns only that set's members.

Changing the Search filter never changes set membership. Returning to Search from Browse restores the previous set filter with the rest of the Search state. If that set was deleted in the meantime, the filter returns to **Ignore**, and **Current results** on the Sets tab also ignores the deleted set.

Saved filters

A set can store one optional filter. Choose **Save current Search** to save the current Search conditions, or **Edit filter** to build conditions directly. The editor supports indexed text, folders, file type, capture date, marked state, and named metadata fields. Combine conditions with nested **All** and **Any** groups. A saved filter is not applied automatically and does not change membership until you choose **Add matches** or **Replace members**.

- **Add matches** adds matching images and keeps existing members.
- **Replace members** makes membership equal to the filter matches. Confirm if the filter matches no images before the set is emptied.

Reapplying checks the whole active database, independent of Search filters and pagination. Long operations show progress and can be canceled; changes already completed are not rolled back.

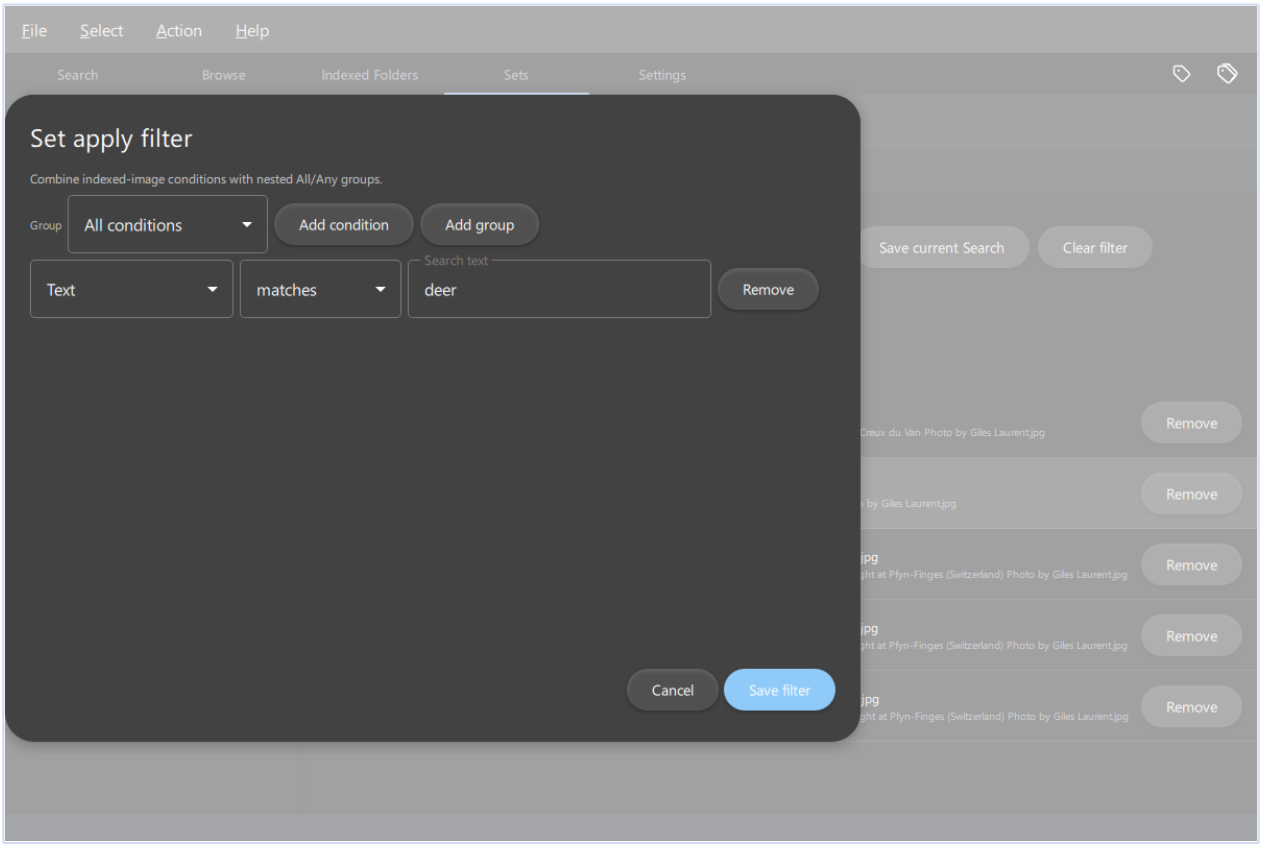


Photo: Wild Red Deer Switzerland, © Giles Laurent, License CC BY-SA 4.0. Visible in the background; screenshot composite licensed CC BY-SA 4.0.

Actions and tagging in Sets

On the **Sets** tab, the **Action** menu targets the selected set, independent of Search filters. Current-results derivative generation uses all set members; marked-image actions use only marked members of that set. JSON export uses marked set members when any are marked, otherwise all members. Mark images from Search or Browse before using a marked-image action in Sets.

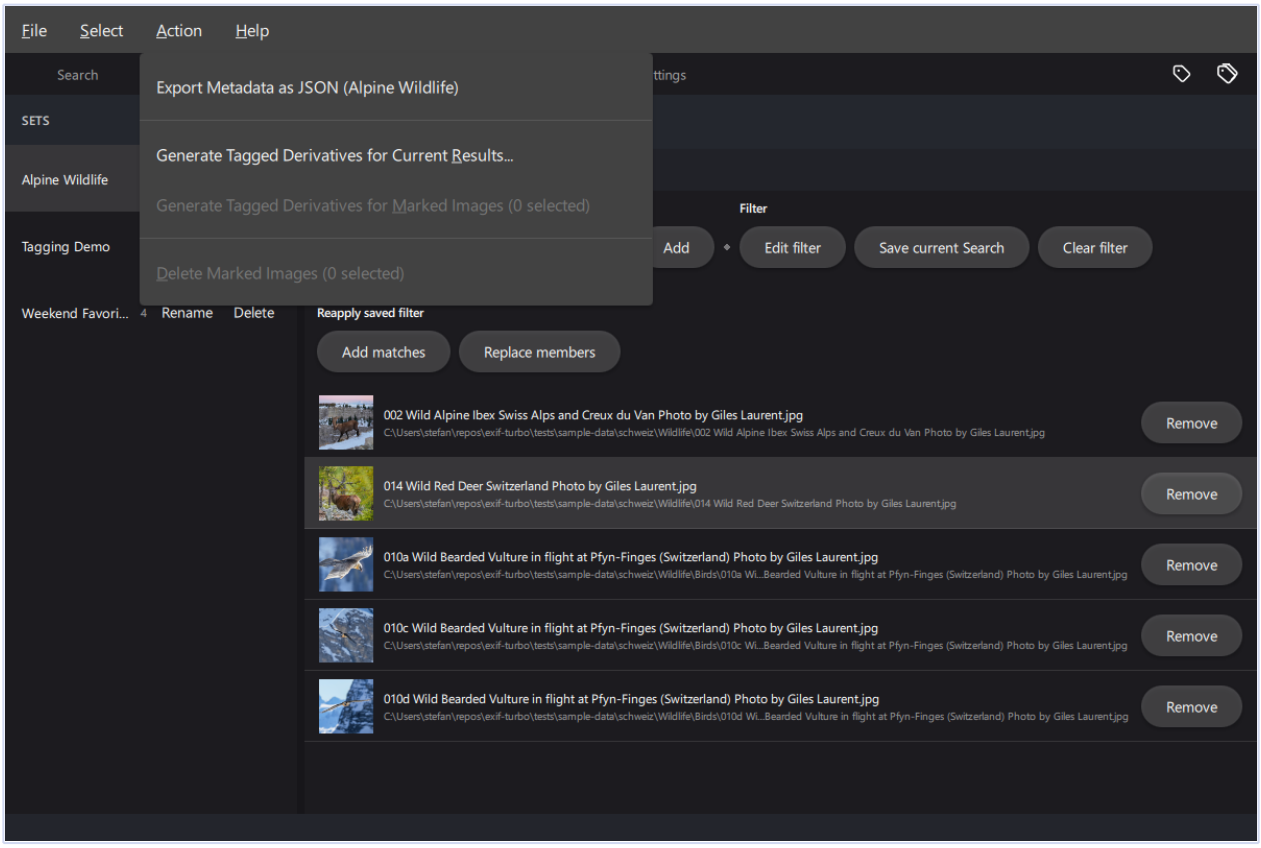


Photo: Wild Red Deer Switzerland, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

The tagging drawer and **Copy tags and descriptions** panel are also available in Sets. Select a member first; double-clicking a member selects it and opens the tagging drawer. See Custom Tags and Derivatives for single-image tagging, set copy targets, and derivative actions.

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9. Viewing Metadata and EXIF Tags

Selecting any image in the result list populates three panels:

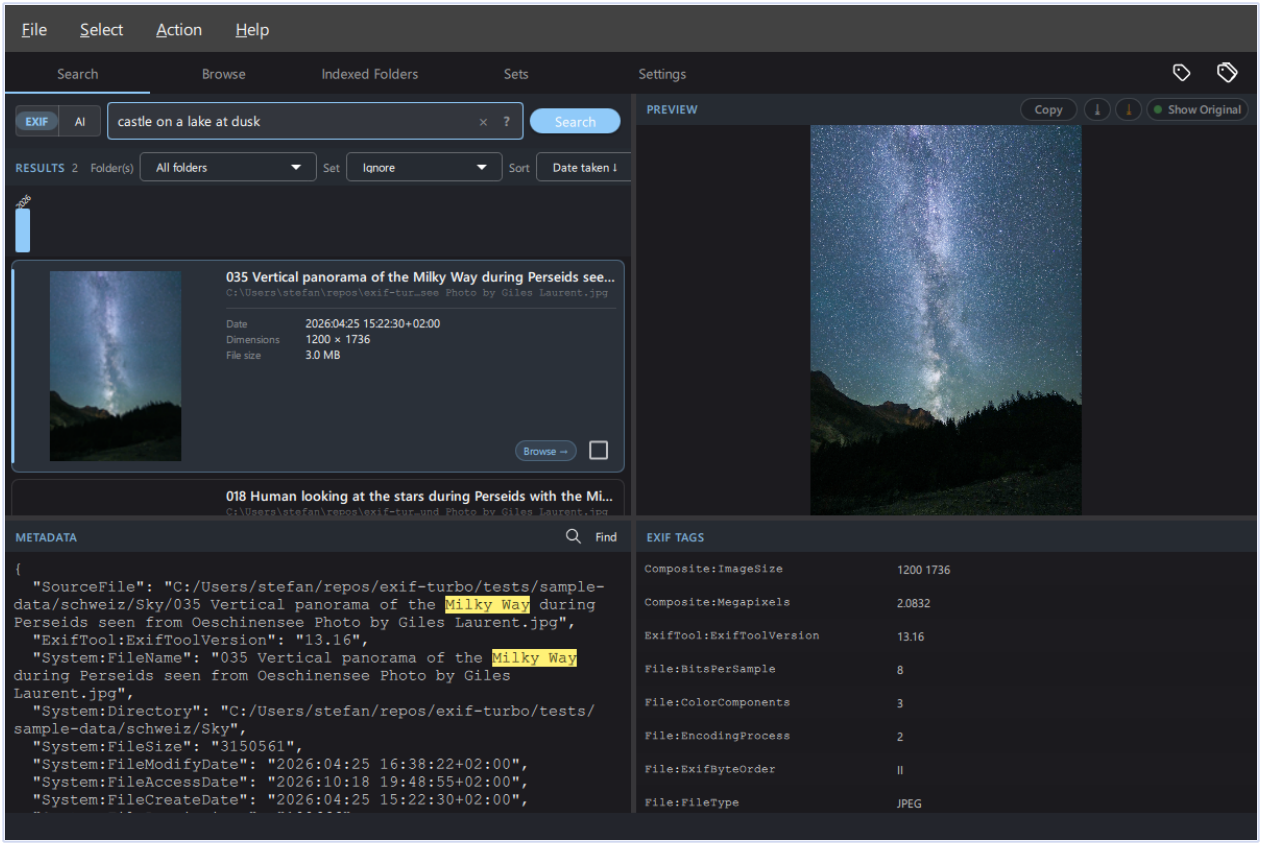


Photo: Vertical panorama of the Milky Way during Perseids, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

Preview

The right pane shows a full-resolution preview of the selected image scaled to fit the available space. For RAW files (CR2, CR3, NEF, ARW, DNG, etc.) the embedded preview JPEG is used. A cached thumbnail is shown immediately as a low-resolution placeholder while the full image is loading; the full image fades in once it has been decoded.

Show Preview / Show Original

When the selected image has a cached preview JPEG, a small pill toggle appears at the right edge of the **PREVIEW** header bar. The label tells you what clicking it will do:

- **Show Original** (with a green dot) — you are currently looking at the cached preview; click to load the full-resolution source file.
- **Show Preview** (with an orange dot) — you are currently looking at the full-resolution source; click to switch back to the cached preview.

The pill is useful for RAW files: the cached preview decodes instantly and is sufficient for most viewing, but switching to **Show Original** loads the full-resolution image when you want to zoom in for detail. The toggle is hidden when no cached preview is available for the selected image.

While a full-resolution original is loading, a small **"Loading original…"** overlay with a spinner appears over the preview area so you know the image is being decoded (large RAW files may take a few seconds).

> The **Show Preview / Show Original** toggle is available in both the **Search** and **Browse** tabs.

Copying the preview image

You can copy the currently displayed preview image to the system clipboard in two ways:

- Click the **Copy** pill button in the **PREVIEW** header bar toolbar (the leftmost of the pill buttons).
- **Right-click** anywhere on the preview image and choose **Copy Image to Clipboard** from the context menu.

Both actions are available in the **Search** and **Browse** tabs. A brief toast notification at the bottom of the window confirms that the image has been placed on the clipboard. If rendering fails for any reason, the file path is copied as plain text instead.

Saving the preview image

A **Save Preview As** button (white ■) appears in the **PREVIEW** header bar toolbar — the second pill from the left — when an image is selected. Clicking it opens a native **Save File** dialog with the filename pre-filled as `<stem>_preview.jpg`, where `<stem>` is the original file's base name without its extension. The available save formats are:

- **JPEG** (`.jpg` / `.jpeg`)
- **PNG** (`.png`)

The saved image is the currently displayed preview. If the **Show Original** toggle is active, the high-resolution source is used; otherwise the cached preview JPEG is saved.

A brief toast notification **"Preview saved"** confirms the export.

You can also trigger this action by right-clicking anywhere on the preview image and choosing **Save Preview As…** from the context menu.

Saving the original file

A **Save Original As** button (orange ■) appears in the **PREVIEW** header bar toolbar — the third pill from the left — when an image is selected. Clicking it opens a native **Save File** dialog with the original filename pre-filled as the suggested name. The source file is copied byte-for-byte to the chosen destination — no re-encoding takes place.

A brief toast notification **"Original saved"** confirms the copy.

You can also trigger this action by right-clicking anywhere on the preview image and choosing **Save Original As…** from the context menu.

Recreate Thumbnail / Recreate Preview

The preview context menu (right-click on the preview image) also offers two rebuild actions, useful when a cached thumbnail or preview ever looks wrong (for example a video frame that was extracted before the rotation fix):

- **Recreate Thumbnail** — deletes the cached thumbnail for the selected image (including any `.skip` sentinel that previously marked it as unthumbnailable) and re-queues thumbnail generation. The thumbnail in the left-hand result grid refreshes automatically as soon as the new file is written — you do not need to switch images or restart the app.
- **Recreate Preview** — deletes the cached preview JPEG for the selected image. The preview pane re-renders the image immediately on next display.

Both actions only affect the selected image; other cached thumbnails and previews are left untouched.

Zooming and panning

The preview supports cursor-anchored zoom and drag-to-pan:

Input	Action
Ctrl + scroll wheel (mouse)	Zoom in / out, anchored to the cursor position
Ctrl + two-finger scroll (touchpad)	Zoom in / out, anchored to the cursor position
Touchpad — pinch	Zoom in / out, anchored to the pinch centroid
Click and drag / **two-finger drag**	Pan the zoomed image in any direction
Plain scroll wheel / two-finger scroll	Pan vertically when zoomed in
Double-click / **double-tap**	Reset zoom to fit

Zoom is capped at **16×**. A badge in the bottom-right corner of the preview shows the current zoom level when it is above 1×. Selecting a new image always resets the zoom to fit.

Metadata panel (bottom-left)

Displays the metadata for the selected image as formatted, indented JSON. Click the **Find** button in the panel header (or press **Ctrl+F**) to open an inline search bar and find any tag value. Press **F3** (or click the ▼ / ▲ arrows) to jump through all matches. Search terms from the main search bar are

highlighted automatically.

GPS location bar

When the selected image contains GPS coordinates in its EXIF metadata, a thin bar appears directly below the **Find** bar. It shows a ■ icon, a **"GPS location —"** label, and three links you can click to view the location on a map:

Link	Service
OpenStreetMap	Opens the coordinates on OpenStreetMap at zoom level 14
Google Maps	Opens the coordinates in Google Maps
GeoHack	Opens the Wikimedia GeoHack coordinate hub — lists nearby Wikipedia articles and dozens of map services

The bar is hidden when the selected image has no GPS data, or when nothing is selected. Hovering over a link shows the full URL in a tooltip.

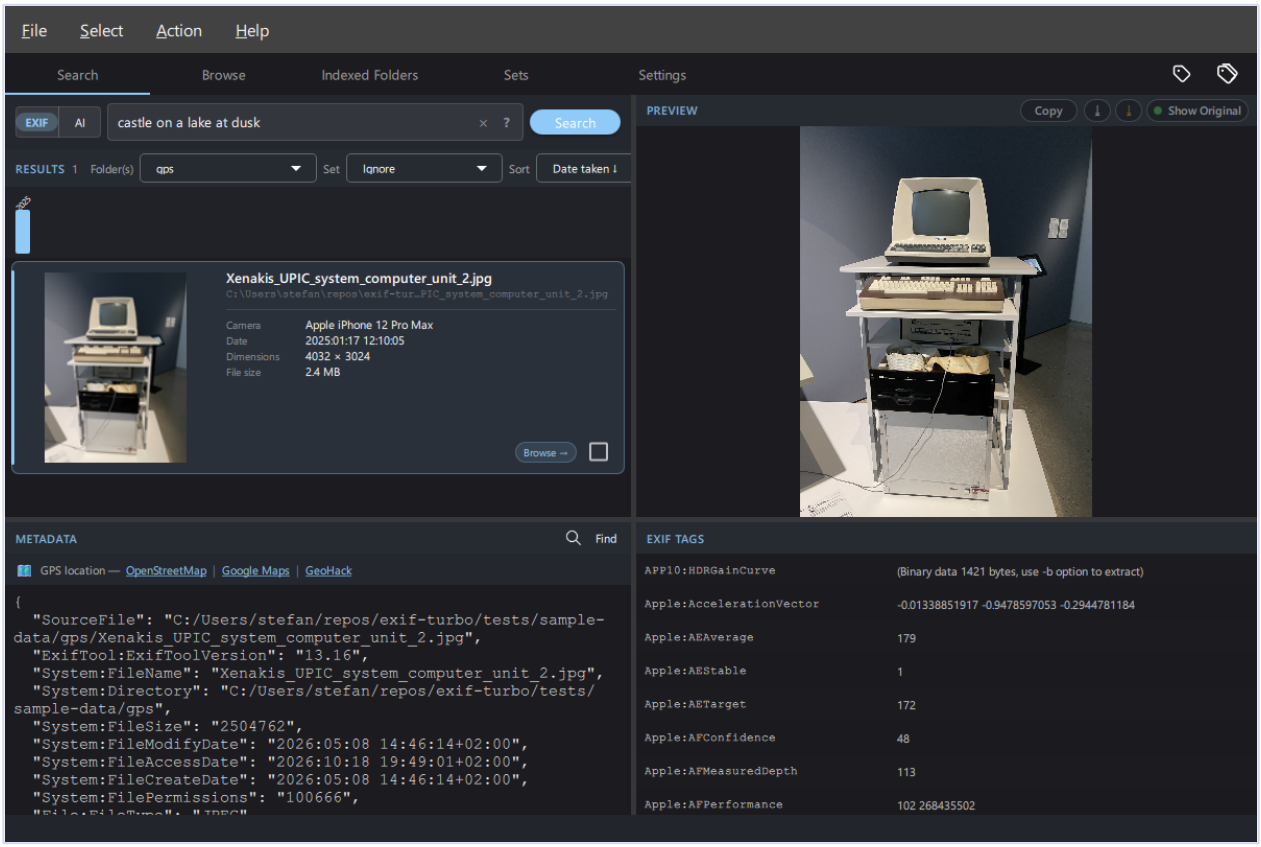


Photo: Xenakis UPIC system computer unit by 1904.CC (Manuel Schmalstieg), CC0 1.0, via Wikimedia Commons. Scaled/cropped in the UI; attribution is voluntary.

EXIF Tags panel (bottom-right)

Displays the same metadata as a clean two-column list, sorted alphabetically. Hover over a truncated tag or value to see the full text in a tooltip.

You can drag the divider between the two bottom panels to adjust the split. By default they start at **50 % / 50 %**.

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10. Marking Images & Bulk Actions

Every result card has a **checkbox** in its bottom-right corner. Tick it to *mark* that image — marks persist across searches, tab switches, and app restarts (the state is stored in the encrypted database). The same checkbox is present on the cards in the **Browse** tab, so you can mark images while browsing as well as while searching. Marked images are the input for the bulk actions in the **Action** menu.

The menu bar exposes two menus dedicated to marking and bulk actions.

Select menu

Action	What it does
Select All	Marks every image matching the current filters (search query, format chip, folder filter, marked-only toggle). Runs on a background thread in batches so the UI stays responsive.
Deselect All	Unmarks every image matching the current filters.
Invert Selection	Flips the marked state of every image matching the current filters.

Select Images Without Thumbnail	Marks every matching image whose thumbnail has not been cached on disk yet. Images the thumbnailer has permanently given up on (oversized files or decoder errors, recorded as a `.skip` sentinel) are also marked, so you can act on them — for example by exporting the metadata or deleting the offending files. Like the other Select actions, it respects the active search query, format chip, folder filter, and marked-only toggle.

All four actions show the **bulk-op progress overlay** with a live `X / Y` counter and a **Cancel** button.

The same modal overlay is also reused for the two long-running maintenance actions — **Remove Folder** (Indexed Folders tab) and **Reset Database** (Settings tab). For those, the overlay adds a sub-step detail line describing the current phase, and for phases that must not be interrupted (such as the database vacuum during a reset) it hides the **Cancel** button and shows a *"This step cannot be canceled…"* notice instead.

Action menu

On Search and Browse, the Action menu uses the marked images or the complete current result scope, as described below. On Sets, it is restricted to the selected set: marked-image actions use only marked members, and current-results actions use all set members.

Action	What it does
Export Metadata as JSON…	Writes the EXIF metadata of every marked image to a JSON file you choose via a Save dialog. The export honours the current **Sort** order (date taken, filename, path, or size). When nothing is marked, it exports all current results. On Sets, it exports marked members when present, otherwise all members of the selected set. The on-disk layout is controlled by **Settings → JSON Export Formatting** (see *Settings*).
Generate Tagged Derivatives for Current Results…	Opens the derivative export dialog for every image matching the current Search filters, including unloaded pages. On Sets, it processes every member of the selected set.

Generate Tagged Derivatives for Marked Images…	Opens the derivative export dialog for marked images. On Sets, only marked members of the selected set are used. Disabled when the current scope has no marked images.
Delete Marked Images…	Permanently deletes marked images **from disk** and removes their rows from the index. On Sets, only marked members of the selected set are targeted. Cached thumbnails (`.png` / `.enc`), `.skip` sentinels, and rendered previews (`.jpg` / `.jpg.enc`) are also cleaned up. Disabled when the current scope has nothing marked.

Long-running actions show progress and support cancellation. Any completed derivatives remain in the target folder, and any deletions already made stay on disk and in the index; the database is never out of sync with the file system.

Confirming a delete

Because **Delete Marked Images…** is irreversible, the confirmation dialog requires an extra step:

1. The dialog states how many files will be deleted (e.g. *"Permanently delete 12 marked image file(s) from disk and remove them from the index?"*).
2. A text field asks you to **type the exact count** — for example `12` — into the box labelled *"To confirm, type the number 12 below"*.
3. The **Yes** button stays disabled until the typed number matches the expected count exactly. Pressing **Enter** in the field also confirms the action when the count matches.
4. **Cancel** (or closing the dialog) aborts without deleting anything.

After completion the status bar reports the outcome, e.g. **"Deleted 12 image(s)."** When some files were already gone or could not be removed (for example because of file-system permissions) extra clauses are appended: **"3 were already missing."** and/or **"1 could not be deleted."**

> **There is no undo.** Files are removed using the operating system's regular delete call — they do *not* go to the Recycle Bin / Trash. Make sure the marked set is correct before confirming.

Tip — auto-sized menus

The Select and Action menu popups are sized to fit their longest item, so dynamic labels such as *"Delete Marked Images… (1234 selected)"* are always shown in full and never truncated.

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11. Custom Tags and Derivatives

Tagging is optional for each database. Enable **Custom Tags** in Settings, then open the drawer from Search, Browse, or Sets with the tag button or **Ctrl+T**. Add your own labels, append or replace a custom description, reuse remembered labels, and control which embedded keywords are included in derivative copies. There are no vocabulary lookups or AI tag suggestions. AI image search and AI-Scan are separate features.

In Settings, **Auto-Repair Tags for Derivatives** is enabled by default and can be changed per database. It repairs a known malformed JPEG XMP header before writing derivative metadata. **Write Description to EXIF XPComment** is also enabled by default; turn it off to write the final derivative description only to XMP.

Sidecars and search

The first tagging change for `photo.jpg` creates `photo.jpg.sidecar.json` beside the image. Sidecars are deterministic UTF-8 JSON and the authoritative store for custom tags, descriptions, and derivative exclusions. They are plain text: SQLCipher database encryption does **not** encrypt them, so they inherit the source folder's permissions and backup policy. Tagging never changes the original image's bytes or timestamp.

Custom labels are copied into the encrypted database's reusable tag catalog and FTS5 cache. Custom descriptions are also cached and searchable using the normal EXIF query box and syntax. Folder scans and **Refresh Tags** synchronize new, changed, and deleted sidecars without re-extracting EXIF. Malformed sidecars are reported and left untouched.

On the first unlock after upgrading, EXIF Turbo removes old accepted vocabulary tags from sidecars and purges their database caches. Custom tags, embedded-tag exclusions, unknown sidecar fields, and original images are preserved. A malformed or externally changed sidecar is left intact and cleanup retries on a later unlock.

Move or rename a sidecar together with its image. EXIF Turbo does not infer an external rename. Removing an image or indexed folder clears only database cache rows; it does not delete a sidecar. Deleting a sidecar removes its custom tags and custom description from FTS after the next synchronization.

Tagging drawer

From the **Search**, **Browse**, or **Sets** tab, click the tag button at the upper right or press **Ctrl+T**. In Sets, select an image in the member list first, or double-click a member to select it and open tagging. This does not change the focused Search image. The non-modal drawer has two independently scrollable columns: **Existing image tags** followed by **Custom tags** on the left; **Existing embedded description** followed by **Custom description** on the right. **Final derivative tags** and **Final**

derivative description remain visible side by side at the bottom, below both scrollable columns. When the Search or Browse image list has focus, arrow keys continue to select images and update the preview, metadata, and tagging panel while it is open.

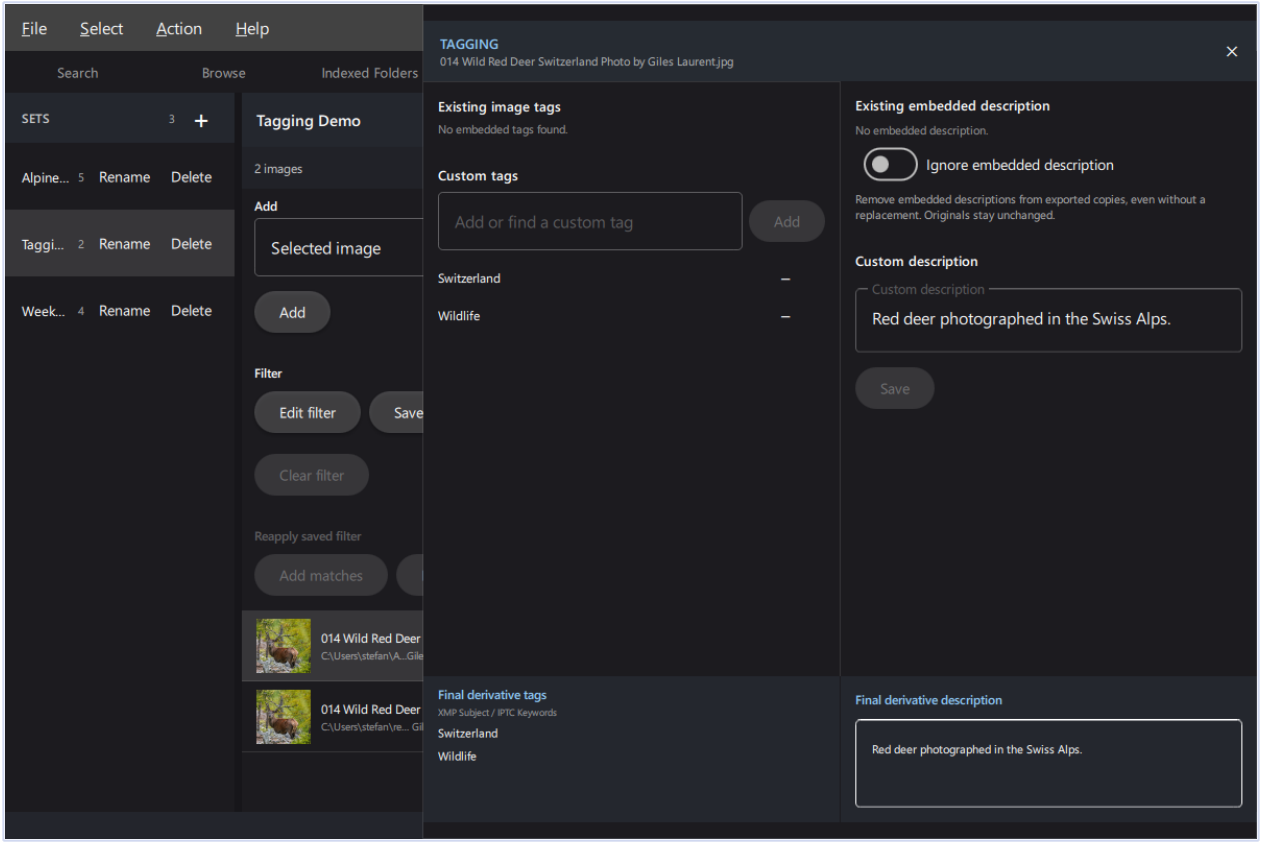


Photo: Wild Red Deer Switzerland, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

It contains these controls:

- **Existing image keywords** lists keywords already embedded in the original image's XMP or IPTC metadata. Select individual tags to exclude them from derivatives, or select **Ignore all existing tags**. These choices are saved in the image sidecar. They never change the source image.
- **Custom tags** accepts a new label with **Enter** or **Add**. Previously used labels are remembered for the current database and appear as suggestions; click one to reuse the same spelling. Removing a custom tag from an image does not remove it from the remembered list. Custom tags do not require AI.
- **Tags on current image** lists custom tags assigned to the focused image. Use the remove button to remove a tag from that image; the remembered suggestion remains available.
- **Existing embedded description** groups the original description and its **Ignore embedded description** switch. **Custom description** is a separate section below **Custom tags**, with one editor showing text saved in the sidecar and a **Save** button. Embedded text includes EXIF ImageDescription (including `IFD0:ImageDescription`), IPTC captions, XMP descriptions and language variants, and supported PNG/QuickTime descriptions. Carriage-return line breaks are

displayed as separate lines. The **Ignore embedded description** switch strikes out embedded text as a preview of its exclusion from exported copies; it remains visible for inspection. An empty custom editor means that no sidecar description was saved. Long embedded descriptions and custom descriptions have their own vertical scrollbars so the entire text stays accessible. The custom description is edited using a multiline draft. **Save** replaces the entire custom description; saving empty text clears it. **Enter** saves, while **Shift+Enter** inserts a newline. Text is never saved automatically on focus loss. The MCP interface retains explicit append and replace modes. Unsaved drafts are protected when the selected image changes; an external or MCP update does not silently overwrite a draft. Custom descriptions are separate from embedded EXIF/IPTC/XMP descriptions, which remain untouched.

- **Ignore embedded description** saves an independent sidecar flag. When enabled, exported copies have their embedded XMP/IPTC/EXIF, PNG, and QuickTime description fields removed and verified, even if no custom description is supplied. Originals and source metadata search stay unchanged. Custom text and unsaved description drafts are not cleared. Disable the switch to retain embedded descriptions in subsequent exports. Copy Tags does not copy this flag. If removal fails, the exporter reports an error and does not finalize that copy.
- **Final derivative keywords** previews the deduplicated custom tags and included embedded keywords that a derivative will receive. MCP clients can control these exclusions with `set_image_embedded_tag_excluded(image_id, path, tag, excluded)` for one keyword or `set_image_all_embedded_tags_excluded(image_id, path, excluded)` for all embedded keywords. Use `excluded=False` to restore inclusion. Individual exclusions remain saved when ignore-all is toggled; restoring one keyword does not override ignore-all. Both settings are returned in MCP image reads and set-image listings. Originals and custom tags remain untouched.
- **Final derivative description**, at the bottom beside the keyword preview, shows the included embedded description followed by the saved custom description, separated by one blank line. Ignoring the embedded description leaves only custom text; with neither, the preview shows an empty-state message. Unsaved drafts are not included. The preview is scrollable. When custom text is present, export clears old embedded description fields and writes/verifies the combined text in XMP-dc Description on the copy. With no custom text, embedded descriptions are retained unchanged unless ignored. Original files are never modified.

Copy panel

Click the **multiple-tags icon** beside the tagging icon in Search, Browse, or Sets to open **Copy tags and descriptions**. This separate non-modal panel replaces the tagging drawer while open; the original tagging drawer remains editable when reopened.

The copy panel shows the selected source's **Custom tags** and saved **Custom description** as read-only previews on the left, with copy controls on the right. **Final derivative tags** and **Final derivative description** remain read-only at the bottom. Unsaved description drafts are neither displayed as saved text nor copied. These previews follow the selected image. To edit the source,

switch back to the normal tagging drawer.

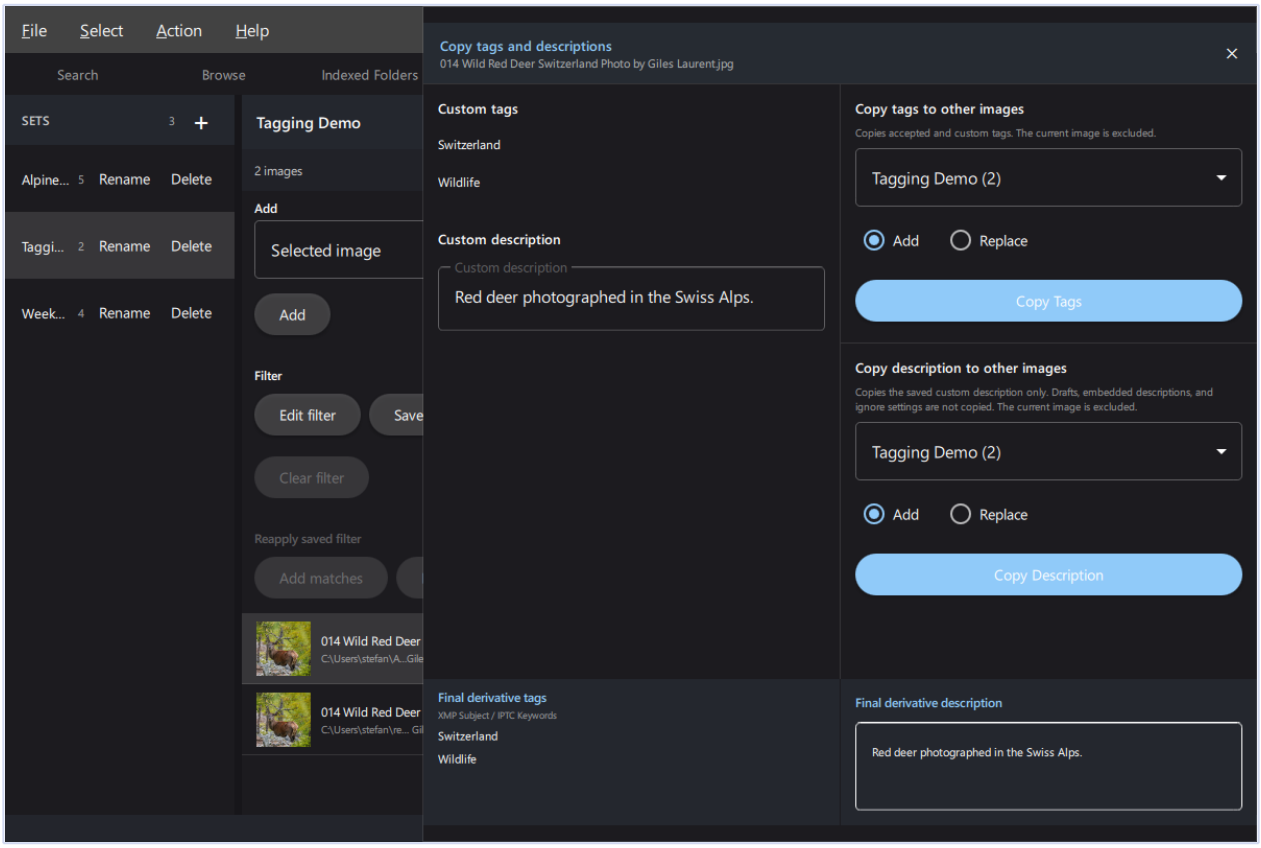


Photo: Wild Red Deer Switzerland, © Giles Laurent, License CC BY-SA 4.0. Scaled/cropped in the UI; screenshot composite licensed CC BY-SA 4.0.

- **Copy Tags** copies custom tags and embedded-keyword exclusions to **Marked images**, **Current search results** (including unloaded pages), or **Current folder** while browsing. Individual exclusions transfer only when the same embedded keyword exists on the target. **Add** merges tags and exclusions; **Replace** substitutes the target's custom tags and exclusion settings and requires confirmation. The focused source image is always excluded. Progress and cancellation are available while the copy runs. On Sets, the default targets are the current set and its marked members. Every named set is also available as a destination from Search, Browse, or Sets. Choosing another set copies only to that destination's members, regardless of current Search filters. The focused source image is excluded even if it belongs to the destination set. Copy Tags does not copy custom descriptions.
- **Copy Description**, below Copy Tags in the copy panel, copies the saved custom description using the same target choices. **Add** appends it as a new paragraph, separated by one blank line; it is disabled when the source description is empty. **Replace** overwrites each target's custom description, or clears it when the source is empty. Replacement and clearing require confirmation. Embedded descriptions, ignore settings, tags, and unsaved drafts are not copied. The source image is excluded. Progress, cancellation, and per-image error reporting work like Copy Tags; completed writes remain if canceled.

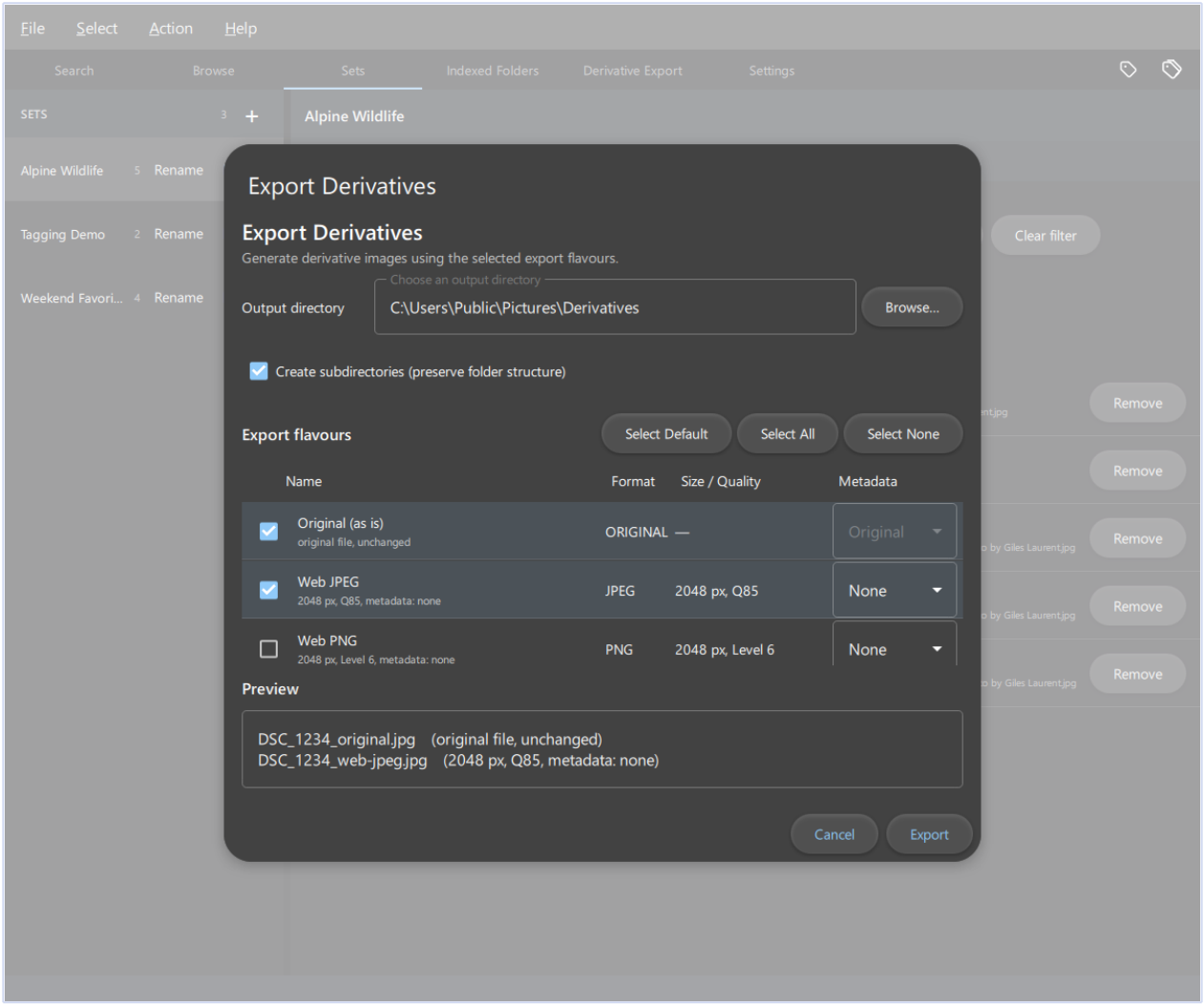
Derivative generation starts from the **Action** menu. The final derivative sections at the bottom are read-only previews.

Marks and bulk scopes

Custom-tag add and remove edit only the focused image. The copy panel's **Copy Tags** action can use the persistent marked set as a bulk target, while marks also remain available as a derivative scope. Press **Space** to toggle the focused image's mark, then choose **Marked images** in Copy Tags or use **Action** → **Generate Tagged Derivatives for Marked Images…**.

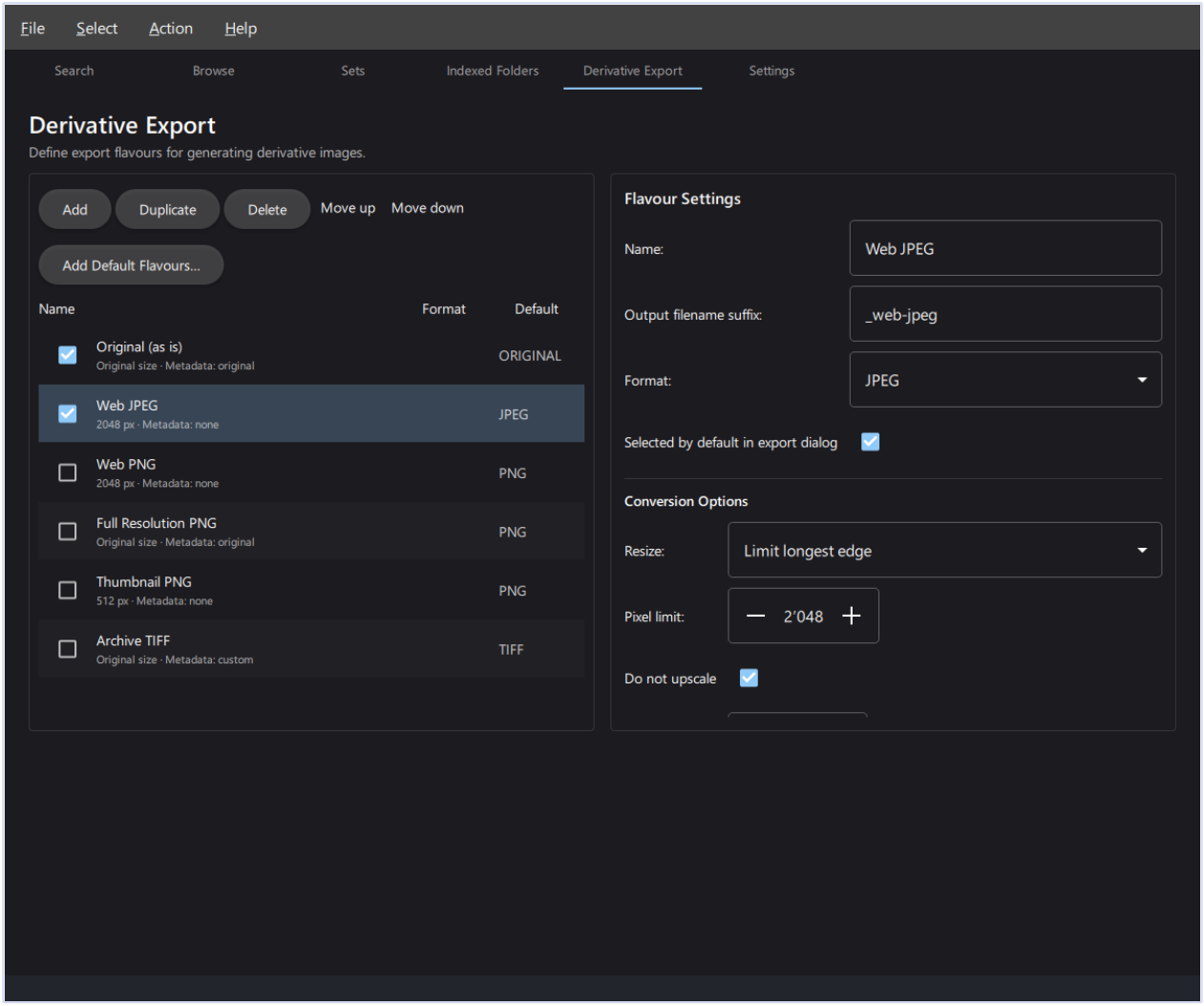
Derivative export flavours

Use **Action** → **Generate Tagged Derivatives for Current Results…** to process the complete current result set, including pages not yet loaded into the view, or **Action** → **Generate Tagged Derivatives for Marked Images…** to process the marked set. The dialog lets you choose an output directory, preserve source subfolders, and select one or more configured flavours. Its default selections come from the **Derivative Export** tab. **Select Default**, **Select All**, and **Select None** affect only the current dialog. Per-flavour metadata overrides are also temporary and reset to the saved flavour value when the dialog is opened again. The chosen output root must be outside every indexed source root.



The preview shows the names and settings that will be used before export begins.

Flavour presets are stored with the active database settings. Each preset has its own format, filename suffix, resize limit, encoder options, metadata mode, and default-selection flag. The initial suggestions are Original (as is), Web JPEG, Web PNG, Full Resolution PNG, Thumbnail PNG, and Archive TIFF. Multiple flavours may use the same output format.



The selected Web JPEG preset uses the `_web-jpeg` suffix, JPEG format, a 2,048-pixel longest-edge limit, and no upscaling.

Original (as is) copies the source bytes without re-encoding or metadata writes, keeps the original extension, and may add a destination filename suffix. Converted flavours support JPEG quality, PNG compression, TIFF None, LZW or ZIP compression, and aspect-preserving size limits without upscaling by default. A video converted to an image produces one representative frame; the Original flavour copies the complete video unchanged.

Each converted flavour uses one of three metadata modes:

- **None** strips removable metadata, including embedded profiles and DPI. Required format/container fields remain.
- **Original** preserves compatible source metadata and does not add exif-turbo tags or descriptions.
- **Custom** applies the existing embedded keyword and description merge, custom tags, and per-image exclusions.

The exporter:

- preserves relative source folders when that option is selected;

- adds collision-safe top-level labels when marks span multiple indexed roots;
- skips existing destination files without overwriting and reports failures per source and flavour;
- stages outputs before finalization and leaves source files unchanged;
- uses ImageMagick for converted images and FFmpeg for video-frame extraction;
- uses the Original flavour for byte-for-byte copies, bypassing the metadata writer;
- merges existing embedded keywords with custom tags, removes case-insensitive duplicates, and writes the result to **XMP Subject** and **IPTC Keywords** for Custom metadata outputs;
- reads the copied file immediately before writing so non-excluded keywords changed since the last index scan are preserved, then verifies both fields with ExifTool;
- writes the final derivative description to **XMP Description** and, when the setting is enabled, **EXIF XPComment**, and also writes it to **IPTC Caption**;
- preserves the full caption in IPTC even when it exceeds the 2,000-byte IPTC-IIM limit; such over-limit IPTC values may not be supported by every metadata reader;
- combines distinct embedded XMP descriptions and IPTC captions into one source description, avoids duplicating identical values, and honors the per-image embedded-description exclusion;
- keeps the converted output if keyword or description metadata cannot be written or verified; those derivatives may have incomplete metadata, and details, including the source, flavour, and destination, are saved in a `derivative-export-errors*.json` report in the output root;
- reports file-copy, conversion, and finalization failures, and removes their incomplete temporary copies;
- never copies sidecars into the derivative tree.

Originals are explicitly forbidden as metadata-write targets. Some metadata fields cannot be preserved across formats, and not every source can be decoded by every installed encoder. Such limitations are reported per output. The source remains unchanged.

Lifecycle and reset

Disabling tagging hides the workbench but does not delete sidecars or custom tags. Closing the app requests cancellation of running workers; completed sidecar writes remain in place.

Reset Database clears indexed images, marks, indexed folders, and thumbnail and preview caches. It deliberately does not traverse source folders to delete adjacent sidecars. The separate image AI index files are not explicitly deleted by reset; use **AI Full Scan** after rebuilding the image index when a clean semantic index is required. Re-add and scan folders to synchronize sidecars.

---

12. Settings

Click the **Settings** tab to configure application behaviour.

Custom Tagging

The custom-tagging switch enables or disables the tagging drawer for this database. It does not affect AI image search or AI-Scan. Previously used custom labels remain available as suggestions while tagging is enabled.

MCP Interface

The optional MCP server runs inside the desktop application and is disabled by default. In **Settings** → **MCP Interface**, enable **Enable MCP server for this database**. The default host is `127.0.0.1` and the default port is `8765`. When the server is running, copy the displayed endpoint into an MCP client configured for Streamable HTTP. The default endpoint is `http://127.0.0.1:8765/mcp`.

Keep the loopback host unless you intentionally need access from another device. Changing the host may expose the interface to other machines that can reach the computer; the MCP endpoint does not have a separate login. The server operates on the currently open, unlocked database and stops with the desktop app. MCP write tools also require **Custom Tags** to be enabled.

MCP clients can inspect the selected image, list sets, read paginated set members, and apply custom tags or descriptions to indexed images. They can also set embedded-keyword and description exclusions for derivative exports. MCP does not create, rename, delete, or change set membership. Batch updates use image IDs and paths from the active database and report per-image outcomes.

Read the current selection or a set

`get_selected_image` returns the image selected in the tagging drawer, including its image ID, path, custom tags, custom description, metadata, and an available JPEG preview. In Search or Browse it follows that tab's selection. In Sets, a set member must be selected; it does not fall back to an unrelated Search or Browse selection. Set `include_preview=False` to omit the inline preview.

Use `list_image_sets` to get set IDs, names, member counts, and saved-filter availability. Then read a set in pages:

```
list_image_sets()
get_image_set_images(set_id=1, include_previews=True)
```

Each page includes image IDs and paths, metadata, custom tags, descriptions, and preview resource URIs. Inline previews default to 5 images and are limited to 5. Without inline previews, pages default to 25 and can contain up to 50 images. Pass `next_offset` as `offset` to fetch the next page; stop when it is `null`.

Update tags and descriptions

`add_custom_tags` adds tags to the image selected in the tagging drawer. It does not update every member of an active set. For explicit targets, use `set_custom_tags_batch` with `image_id`, `path`, `tags`, and optional `mode` (`append` by default, or `replace`):

```
set_custom_tags_batch(updates=[
 {"image_id": 42, "path": "/photos/alpine-lake.jpg",
  "tags": ["alpine lake", "summer"], "mode": "append"},
])
```

Tag batches accept up to 100 images. Each result reports success or error and, on success, the resulting custom tags. Use matching image IDs and paths from `get_selected_image` or `get_image_set_images`; both must identify the same image in the active, unlocked database. Successful tag updates are recorded at INFO level with the image ID, path, mode, and requested and resulting tags.

Use `set_image_description` to append or replace one image's custom description. The mode is required; append adds a paragraph separated by a blank line, while replace substitutes the full description. An empty append is rejected and an empty replacement clears the custom description. For example:

```
set_image_description(
 image_id=42,
 path="/photos/alpine-lake.jpg",
 description="Photographed shortly before sunset.",
 mode="append",
)
```

`set_image_descriptions_batch` applies the same operations to up to 100 images. Each update must include `image_id`, `path`, `description`, and explicit `mode`; results report per-image success or error, the resulting description, and a `changed` flag. Custom descriptions are stored in adjacent sidecars and the search index. MCP writes never modify original image bytes or embedded metadata.

Control derivative exclusions

These tools change sidecar settings for exported copies only; source image metadata and ordinary metadata search remain unchanged. Tag-exclusion tools require a matching indexed image ID/path, an unlocked database, and **Custom Tags** enabled. Individual embedded-keyword exclusions are case-insensitive and remain saved when ignore-all is toggled. Restoring one keyword does not override ignore-all; clear the all-keywords flag separately.

```
set_image_embedded_tag_excluded(
 image_id=42, path="/photos/alpine-lake.jpg", tag="Family", excluded=True
)
set_image_all_embedded_tags_excluded(
```

```
 image_id=42, path="/photos/alpine-lake.jpg", excluded=True
)
set_image_description_excluded(
 image_id=42, path="/photos/alpine-lake.jpg", excluded=True
)
```

Set `excluded=False` to restore inclusion. Tag-exclusion results include `excluded_embedded_tags`, `exclude_all_embedded_tags`, and `changed`; description exclusion results include `exclude_embedded_description` and `changed`. These settings are also returned by selected-image reads and set-image listings. Ignoring an embedded description does not clear custom description text.

Worker Threads

Controls the number of parallel threads used for indexing and thumbnail generation. Higher values speed up processing on multi-core machines but use more CPU and memory. The default is half the number of detected CPU threads.

Preview Cache Size

Long-edge resolution used by the preview-cache builder. Larger values give sharper detail when zooming but take more disk space and longer to render. Choose a value (in pixels) from the dropdown — the change applies to subsequently built previews; existing cached previews are unaffected until you rebuild them via **Build Previews** on a folder row.

libvips Allowed Extensions

Controls which file extensions may use the native libvips fallback for images that are unusually large or cannot be decoded correctly by Pillow. The default list covers JPEG, PNG, TIFF, WebP, and GIF. Add an extension such as `.bmp` only when you need libvips support for that format; remove an extension to prevent that format from reaching libvips. Normal Pillow decoding remains available.

Changes apply immediately and are stored per database. Operations that libvips marks as untrusted remain blocked regardless of this list; adding an extension does not weaken that mandatory protection.

Indexing Blacklist

A list of file and folder name patterns to skip during indexing. Supports wildcards: `*` matches any sequence of characters, `?` matches a single character. Patterns are matched against individual file or folder names, not full paths.

Examples: `@eaDir`, `*.tmp`, `Thumbs.db`

Changes to the blacklist take effect on the next rescan.

JSON Export Formatting

Controls how the **Export Metadata as JSON…** action writes its output.

- **Pretty-print (indented) JSON** — off by default, which keeps the historical compact layout (the export is a JSON array with each record serialised on its own line). Turn it on to indent every record for easier human reading.
- **Indent style** — when pretty-printing is on, choose **Spaces** or **Tabs**.
- **Indent size** — when the style is **Spaces**, choose how many spaces make up one indentation level (2, 4, or 8).

Whatever the format, the file is always valid JSON that round-trips back to the same records. The setting is stored per database and applies to the next export.

Theme

Choose between **system** (follows OS dark/light mode), **light**, or **dark**. The theme changes immediately.

Language

Select the application display language from the dropdown. A restart is required for that change to take full effect.

ExifTool

Click **Check** to verify that ExifTool is installed and available on your `PATH`. A colour-coded badge appears next to the button:

- **Green dot + "Found — ExifTool \<version>"** — ExifTool was found and the version is shown.
- **Red dot + "Not found"** — ExifTool is missing. A download link (exiftool.org) and a restart hint are shown below the badge.

The badge is also populated automatically each time you unlock the database, so you do not need to click **Check** after every launch.

Change Password

Click **Change Password…** to re-encrypt the SQLCipher database under a new password. The dialog asks for:

- **Current password** — must match the password the database is currently encrypted with.

- **New password** — your new passphrase. Cannot be empty and must differ from the current one.
- **Confirm new password** — must match the new password exactly.

Click **Change Password** to apply. A busy indicator and the message **“Changing password… This may take a moment.”** are shown while exif-turbo re-encrypts every page of the database. Existing **thumbnails are preserved** — exif-turbo uses a wrapped-key model, so only the wrapping key (which protects the thumbnail key) is re-encrypted under the new password; the thumbnail cache on disk does not have to be rebuilt.

The AI master key is also re-wrapped; the encrypted vector snapshot is not rewritten. A staged wrapper allows the next verified database unlock to recover an interrupted password change. After a successful change, the old password no longer unlocks the current AI key. Historical backups retain their old passwords.

Once the operation finishes successfully the dialog closes and a confirmation appears. The new password is required the next time you unlock the database.

If the current password is wrong, an inline red error message (“Current password is incorrect.”) is shown and the dialog stays open. The **Change Password…** button is disabled while indexing is in progress and while the database is locked. The operation is also rejected if thumbnail building is running when you submit the dialog. Password changes are also rejected while AI scanning, preview building, or background search/database-reading workers are running.

> **There is no recovery if you forget the new password.** Make sure you remember it (or store it in a password manager) before clicking **Change Password**.

Reset Database

At the bottom of the Settings tab, a divider separates the standard settings from a destructive-action zone.

Click the ■■ **Reset Database…** button (red) to open a confirmation dialog. Click **OK** to confirm. This permanently:

- Deletes all indexed images from the database
- Removes all indexed folder records
- Wipes the thumbnail and preview cache on disk

The database is vacuumed and checkpointed immediately, so the database file shrinks to near-zero on disk straight away.

While the reset runs, the modal **bulk-op progress overlay** (see section 10) covers the window and steps through each phase: *"Clearing preview cache…"* (cancelable, with a live `X / Y` count), *"Deleting index rows…"*, and finally *"Vacuuming database…"*. The vacuum phase cannot be interrupted — the overlay hides its **Cancel** button and shows a *"This step cannot be canceled…"*

notice until it finishes.

> **This action cannot be undone.** After a reset you will need to re-add your folders and run a full rescan to rebuild the index. Adjacent tagging sidecars are not deleted; rescanning synchronizes their custom tags again. Existing image AI vector files are not explicitly deleted; run **AI Full Scan** when you need to rebuild them.

The **Reset Database…** button is disabled while indexing is in progress.

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13. Keyboard Shortcuts

Shortcut	Action
`Enter`	Run search
`Ctrl+F`	Open / close find-in-metadata bar
`Escape`	Close the find-in-metadata bar
`F3`	Jump to next match in metadata
`Shift+F3`	Jump to previous match in metadata
`Ctrl+T`	Open / close the tagging drawer in Search, Browse, or Sets
`Space`	Toggle the mark on the focused Search or Browse image
`↓`	Select the next result (Search tab) / next image (Browse tab)
`↑`	Select the previous result (Search tab) / previous image (Browse tab)
`Page Down`	Jump one page forward in results (Search tab) / Browse image list
`Page Up`	Jump one page backward in results (Search tab) / Browse image list
`Ctrl+Q`	Exit the application (File → Exit)

14. FAQ

Q: Why does the status bar say "Indexing…" even after I switch tabs?
A: The indexer runs in the background across all tabs. The pulsing blue dot in the status bar lets you know it is still working. You can continue searching and browsing while indexing proceeds.

Q: I closed the app mid-scan. Will I lose my progress?
A: No. exif-turbo saves the queue state when it closes. Next time you unlock the database, any interrupted scans are automatically resumed.

Q: The search finds nothing even though I can see files in the folder.
A: The files must be indexed first. Go to the **Indexed Folders** tab, add the folder, and click **Scan**.

Q: Does exif-turbo modify my image files?
A: Tagging, indexing, and derivative generation never modify originals. **Delete Marked Images** is an explicit destructive action that permanently deletes the selected original files after confirmation.

Q: What image formats are supported?
A: JPEG, PNG, TIFF, HEIC, BMP, GIF, and RAW formats: CR2, CR3, NEF, ARW, DNG, ORF, RW2, PEF, RAF, RWL, SRW (and any other format that ExifTool can read). Video files are also indexed: MP4, MOV, AVI, MKV, WMV, M4V, MTS, M2TS, 3GP, WebM, FLV. Thumbnails and previews for video files are extracted via PyAV (FFmpeg) — the embedded thumbnail is used when available, otherwise a frame at 1/3 of the duration. Rotation is applied so portrait phone clips render upright.

Q: Where is the database stored?
A: By default at `~/.exif-turbo/data/index/index.db` on all platforms.

Q: How do I change the database password?
A: Open the **Settings** tab and click **Change Password…**. Enter your current password and a new one — exif-turbo will re-encrypt the database in place. Thumbnails are preserved (no re-indexing required). See Change Password for details.

Q: Thumbnails are not showing / are slow to appear.
A: Thumbnails are generated in a background thread after indexing. Depending on the number of images it may take a few minutes for all thumbnails to be built and cached. They persist across sessions so subsequent launches are fast.

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